STAFFORD'S CONJECTURE ON CYCLICITY OF TORSION MODULES

CHRISTOPHER ALBERT, JOHANNA MOSER, AND MAXIMILIAN PHILIPP

Abstract. We prove that over a Weyl algebra over a field of characteristic zero $A_n(k)$, every nonzero $d \in A_n(k)$ admits $F, R, S \in A_n(k)$ such that $1 = dR + FdS$, a conjecture Stafford proposed in 1978. As a consequence, every finitely generated torsion right $A_n(k)$-module is cyclic. The proof maps d, by a symplectic change of variables, to a normal form $\tilde{d}$ and shows that the cyclic module $Q = A_n(k)/(\tilde{d}A_n(k) + x_1^N \tilde{d}A_n(k))$ vanishes. We first show that right multiplication by x_1 is surjective on Q. A comparison of lengths of localized modules over the associated graded ring then gives $\operatorname{gr} Q = x_1 \operatorname{gr} Q$, so that the characteristic variety of Q is disjoint from the hyperplane $\{x_1 = 0\}$. Finally, Gabber's involutivity theorem together with an argument about conormal directions at infinity from algebraic geometry shows that such a characteristic variety must be empty and therefore, $Q = 0$.

AI comment Review guide. Red text proposes additions or corrections; blue struck text preserves the original wording. **Johanna:** review the proposals as mathematics and prose, and resolve their boxes after accepting or revising them. No Lean installation, code editing, or proof replay is required for her review. **Max:** review the whole paper–Lean correspondence, including every statement, its hypotheses, and the definitions entering the challenge. The technical supplement contains the comparison notes, mathematical accounts, and optional declaration links. The side-by-side review interface links statements to their checked sources.

1. Introduction

Let k be a field of characteristic zero, and let

$$A_n(k) = k\langle x_1, \dots, x_n, \partial_1, \dots, \partial_n \rangle$$

be the nth Weyl algebra, with $[\partial_i, x_j] = \delta_{ij}$ and $[\partial_i, \partial_j] = [x_i, x_j] = 0$. Stafford's Conjecture 3.8 asks whether every nonzero $d \in A_n(k)$ admits $F \in A_n(k)$ with $A_n(k) = dA_n(k) + FdA_n(k)$ [30, p. 438]. Stafford identifies this with stable cyclicity for finitely generated torsion right Weyl-algebra modules [30, pp. 437–438]. Bellamy's survey states the weaker ordinary-cyclicity question for finitely generated torsion $A_n(\mathbb{C})$-modules as open [8, Section 3, Conjecture 3.4]. We prove Stafford's conjecture:

Theorem 1.1. *For every characteristic-zero field k, every $n \geq 0$, and every nonzero $d \in A_n(k)$, there exist $F, R, S \in A_n(k)$ such that*

$$1 = dR + FdS.$$

Moreover, if $n \geq 1$, one may choose $F = \ell^{\deg_B d}$ for a linear Weyl coordinate ℓ, obtained by an invertible linear symplectic change of generators. Explicitly, $\ell = \Phi_g^{-1}(x_1)$ for a linear ω-preserving automorphism g as in Section 3.

Proof sketch: We start in section XXX by defining a map Φ_g sending any chosen d of Bernstein degree N to a monic normal form $\tilde{d}$ with order symbol P using Bernstein and order filtrations of $A_n(k)$. For $\tilde{d}$, we prove in sections XXX that the module $Q = A_n(k)/(\tilde{d}A_n(k) + x_1^N \tilde{d}A_n(k)) = 0$, and find the final form of Theorem 1.1 by applying the inverse map Φ_g^{-1} to $\tilde{d}A_n(k) + x_1^N \tilde{d}A_n(k)$. The main components of the proof are the following: in section XXX, we show that $Q = Qx_1$ using the extended euclidean algorithm, and $\operatorname{gr} Q = x_1 \operatorname{gr} Q$, through two contradictory length estimations of localized modules. In section XXX using mainly results from algebraic geometry, we show that a set W fulfilling certain criteria (Zariski closed over an algebraically closed field of characteristic zero, fibre-conical, involutive, and there exists a polynomial with $P(1, 0 \ldots, 0) \neq 0$ vanishing on W, and W does not contain points with $x_1 = 0$) must always be empty (Theorem 5.6). Finally, we show in section XXX that by defining W to be the characteristic variety of the closed-field pendant to Q, the results of section XXX as well as the definition of P in section XXX ensure that W fulfills all criteria and must thus be empty. We conclude that therefore, also Q itself must be zero, from which Theorem 1.1 follows directly. Lastly, we provide the proof by Stafford [30] that hence, every finitely generated torsion module is stably cyclic (Corollary 6.3).

1.1. **Related literature.** Constructive predecessors concern Stafford's established generator theorems. Hillebrand–Schmale and Leykin give effective two-generator methods, and Leykin also constructs cyclic generators of holonomic modules [18, 21]. Quadrat–Robertz develop module-decomposition and basis algorithms [26, 27]; Caro–Levcovitz and Caro-Tuesta–Levcovitz study differential-operator rings with series coefficients [10, 11]. The constrained identity for arbitrary nonzero d requires the additional quotient-vanishing argument developed here.
The surrounding generator theory includes Stafford's stable-range and noncommutative Forster–Swan results [32, 31]. The generator article has a corrigendum and related exposition [34, 33]. Stafford also constructed stably free Weyl-algebra ideals [35]. OreModules supplies computational methods for modules over Ore algebras [12].

1.2. **Formal verification with Lean.** Theorem 1.1, including the choice $F = \ell^{\deg_B d}$, and Corollary 6.3 are proved in Lean 4 [15] with Mathlib [36] and AlgebraicAnalysis v0.3.3 [2]. The proofs use only Lean's three standard axioms: propositional extensionality (`propext`), the axiom of choice (`Classical.choice`) and soundness of quotient types (`Quot.sound`). Every intermediate result is proved in Lean as well, including Gabber's involutivity theorem for cyclic $A_n(k)$-modules. The formal proof is organized along the same steps as Sections 3–6. The main theorem is registered in Palomar [6],

and the Lean sources are archived [1]. Tooling and the role of AI assistance are described in Appendix A. Section 7 states further consequences proved in Lean and two generalizations, to rings of differential operators on smooth affine varieties and to simple filtered algebras. The supplementary material [7] shows each formalized statement of this paper next to its Lean version and explains the formal proofs in mathematical language. It can be browsed at https://itpplasma.github.io/stafford38-supplementary/.

2. Preliminaries

Throughout, k is a field of characteristic zero and for the n-th Weyl algebra over k, we write

$$A_n(k) = k\langle x_1, \dots, x_n, \partial_1, \dots, \partial_n\rangle, \quad [\partial_i, x_j] = \delta_{ij}, \quad [x_i, x_j] = [\partial_i, \partial_j] = 0.$$

All ideals are *right* ideals and all modules are *right* modules unless the contrary is said explicitly.

Convention 2.1. *We single out the first Weyl pair (x_1, ∂_1) and write*

$$p := \partial_1, \qquad C := A_{n-1}(k)[x_1], \qquad A_n(k) = C[p; \delta],$$

Here $\delta(x_1) = 1$ and δ vanishes on $A_{n-1}(k)$. so that x_1 is central in C, $[p, x_1] = 1$, and the transverse algebra $A_{n-1}(k)$ commutes with x_1 and p. Every element of $A_n(k)$ can uniquely be written as $\sum_j p^j c_j$ with $c_j \in C$.

Definition 2.2 ([Def. 2.6 and 2.9 [28]])**.** *A filtration $F_\bullet$ on $A_n(k)$ is a sequence of linear subspaces*

$$\{0\} = F_{-1} \subseteq F_0 \subseteq F_1 \subseteq \dots \subseteq A_n(k),$$

such that $F_i \cdot F_j \subseteq F_{i+j}$ and $A_n(k) = \bigcup F_j$. The associated graded algebra, also called symbol ring, is defined as

$$\operatorname{gr}^F A_n(k) = \bigoplus_{j=0}^{\infty} F_j/F_{j-1}.$$

Let $a \neq 0 \in A_n(k)$ and let j be the smallest integer, such that $a \in F_j$, called a filtration degree $\deg_F a$*. The class of a in F_j/F_{j-1} is called its* symbol *$\sigma_j(a)$. A symbol, being the class of an element in a single graded piece, is automatically homogeneous of the corresponding degree, i.e. contains only terms of that total degree.*

In the proof, we use two filtrations of $A_n(k)$.

- The *Bernstein filtration* $B_\bullet$ assigns degree 1 to every x_i and every ∂_i. The Bernstein degree of d is written $\deg_B d$.
- The *order filtration* $F_\bullet$ assigns degree 0 to every x_i and degree 1 to every ∂_i; F_m is spanned by the monomials $x^\alpha \partial^\beta$ with $|\beta| \leq m$.

Formal comparison: as spans of ordered PBW words of bounded weight

For both the order filtration and the Bernstein filtration, the associated graded algebra $\mathrm{gr}A_n(k)$ is simply the polynomial ring $k[x,\xi] = k[x_1,\dots,x_n,\xi_1,\dots,\xi_n]$ in $2n$ variables with ξ_i the symbol of ∂_i. In particular, $\mathrm{gr}A_n(k)$ is commutative [28, Prop. 2.10]. Good filtrations and characteristic varieties are treated systematically for example in [9, 19].

2.1. **Cotangent space and symplectic structure.** Let $X = \mathbb{A}^n_k$ be affine n-space over a field k of characteristic zero, with coordinates $x_1,\dots,x_n$. Its *cotangent bundle* T^*X is the space of pairs (y,ξ) consisting of a point $y \in X$ together with a covector ξ at y; since X is affine space, this cotangent space is the same vector space $(k^n)^*$ at every point, and we may identify

$$T^*X \;\cong\; X \times (k^n)^*.$$

We write $\xi_1,\dots,\xi_n$ for the coordinates on $(k^n)^*$ dual to $x_1,\dots,x_n$, so that a covector is $\sum_i \xi_i\,dx_i$, and identify $\mathrm{gr}^B\, A_n(k) = k[x,\xi]$ with the coordinate ring of T^*X. We write $\pi\colon T^*X \to X$ for the projection forgetting the covector, and for $y \in X$ write $T^*_yX = \pi^{-1}(y) \cong (k^n)^*$ for the *fibre* of T^*X over y.
We single out the first coordinate x_1 and define the distinguished hyperplane

$$H := \{x_1 = 0\} = \{(0,x_2,\dots,x_n)\} \subseteq X;$$

its conormal covector is dx_1, i.e. $\xi = (1,0,\dots,0)$.

Definition 2.3. *A filtration of a right A_n-module M compatible with $F_\bullet$ is an exhaustive increasing chain of k-subspaces $M_\bullet$ with $M_iF_j \subseteq M_{i+j}$ and $M_i = 0$ for $i < 0$. It is* good *if the associated graded module* $\mathrm{gr}\,M = \bigoplus_i M_i/M_{i-1}$ *is finitely generated over* $\mathrm{gr}^F\, A_n$.

As an example, the filtration $M_i := qF_i$ of a cyclic module over A_n induced by its generator q is good, since $\mathrm{gr}\,Q$ is generated by the element $\bar{q}$ denoting the class of q in Q_0/Q_{-1}.
For a finitely generated right A_n-module M with a good order filtration $M_\bullet$, the graded module $\mathrm{gr}\,M$ is a finitely generated $k[x,\xi]$-module.

Definition 2.4. *The* characteristic variety *(also called symbol variety) of a finitely generated A_n-module M with a good order filtration $M_\bullet$ is defined by*

$$\mathrm{Char}(M) := V\big(\mathrm{Ann}_{k[x,\xi]}\,\mathrm{gr}\,M\big) \subseteq T^*\mathbb{A}^n = \mathrm{Spec}\,k[x,\xi],$$

$\mathrm{Ann}_{k[x,\xi]}\,\mathrm{gr}\,M = \{f \in k[x,\xi] : f\cdot m = 0\,\forall m \in \mathrm{gr}\,M\}$ *are the* annihilators *of* $\mathrm{gr}\,M$ *over* $k[x,\xi]$*, and V denotes their* vanishing locus*, i.e. the set of points (x,ξ) for which all annihilator polynomials vanish. This set of points lies in the* cotangent bundle $T^*\mathbb{A}^n$ *of the affine n-space* $\mathbb{A}^n$.
$\mathrm{Char}(M)$ *is independent of the choice of the good filtration.*

Formal comparison: for $M = A_n/I$

Definition 2.5. *A bilinear form ω on a k-vector space is* symplectic *if it is antisymmetric and nondegenerate.*
The standard symplectic form on the vector space k^{2n}, with basis $x_1, \dots, x_n$, $\partial_1, \dots, \partial_n$, is determined by the relations

$$\omega(x_i, \partial_j) = \delta_{ij}, \qquad \omega(x_i, x_j) = \omega(\partial_i, \partial_j) = 0,$$

analogously to the Weyl relations. Identifying $T^*X \cong X \times (k^n)^*$ via the coordinates $x_1, \dots, x_n$ and their duals $\xi_1, \dots, \xi_n$, on T^*X this ω takes the familiar form

$$\omega = \sum_{i=1}^{n} dx_i \wedge d\xi_i.$$

The associated Poisson bracket *on $k[x, \xi]$, the coordinate ring of T^*X, is*

$$\{f, g\} = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\frac{\partial g}{\partial \xi_i} - \frac{\partial f}{\partial \xi_i}\frac{\partial g}{\partial x_i} \right).$$

Definition 2.6. *An ideal J is* involutive *if $\{J, J\} \subseteq J$.*

Theorem 2.7 (Gabber [16][29][19, Theorem 2.3.1 and Appendix D])**.** *Let R be a filtered ring such that $\operatorname{gr}(R)$ is a Noetherian commutative algebra over $\mathbb{Q}$. Let M be a finitely generated R module. Let $J = \sqrt{Ann_{\operatorname{gr} R} \operatorname{gr} M}$ be the radical of the annihilators of M. Then J is involutive, i.e.*

$$\{J, J\} \subset J.$$

Formal comparison: proved in Lean; a standard theorem, so its formal proof need not be compared

Theorem 2.8 (Hilbert's Nullstellensatz)**.** *Let K be an algebraically closed field and $J \subseteq K[z_1, \dots, z_N]$ an ideal. Then*

(a) $V(J) = \varnothing \iff 1 \in J$
(b) $I(V(J)) = \sqrt{J}$

3. Mapping to a normal form

Given a fixed nonzero $d \in A_n(k)$ of Bernstein degree N, our first goal is to find a map to replace d by a normal form $\tilde{d}$ that is monic in p, i.e. of the form $\tilde{d} = p^N + \sum_{j<N} p^j a_j$ with coefficients a_j in C.

If $N = 0$, then $d \in B_0 = k$ is a nonzero scalar, hence a unit, and $1 = dR + FdS$ holds trivially with $F = 1 = \ell^0$, $R = d^{-1}$, $S = 0$. We therefore assume $N \geq 1$.
We want to construct a well-behaved automorphism of $\operatorname{gr}^B A_n(k) = k[x, \xi]$ from simpler linear algebra. We proceed in three stages: define an automorphism of the vector space $\mathbb{V} = k^{2n}$, extend it to an automorphism of $A_n(k)$ itself, and observe the induced action this extension leaves on the associated graded ring.

We can identify $\mathbb{V} = k^{2n}$ with the Bernstein-degree-one part B_1/B_0 of $A_n(k)$, i.e. linear combinations of $x_1, \ldots, x_n, \partial_1, \ldots, \partial_n$; the coordinates of a vector are then the coefficients of these generators. By this identification, $\mathbb{V}$ carries the standard symplectic form ω (Def. 2.5).

Lemma 3.1. *Let $(\mathbb{V}, \omega)$ be a symplectic k-space of dimension $2n$ and $0 \neq v \in \mathbb{V}$. Then v is the first vector of a symplectic basis of $\mathbb{V}$; consequently, there exists a k-linear automorphism g preserving ω (i.e. $\omega(gu, gv) = \omega(u, v)$ for all $u, v \in \mathbb{V}$.) with $g(e) = v$ for any prescribed nonzero vector e.*

Proof. Since ω is nondegenerate and $v \neq 0$, the functional $\omega(v, -)$ is not identically zero, so there is $w' \in \mathbb{V}$ with $\omega(v, w') = c \neq 0$; setting $w := w'/c$ gives $\omega(v, w) = 1$.
Let $U = \operatorname{span}(v, w)$. For $0 \neq av + bw \in U$: if $a \neq 0$, $\omega(av + bw, w) = a \neq 0$; if $b \neq 0$, $\omega(av + bw, v) = -b \neq 0$. Either way $av + bw$ pairs nontrivially with something in U, so $\omega|_U$ is nondegenerate.
Since $\omega|_U$ is nondegenerate, $U \cap U^\perp = 0$, and as $\dim U + \dim U^\perp = \dim \mathbb{V}$ for a nondegenerate form, this gives $\mathbb{V} = U \oplus U^\perp$. If $u \in U^\perp$ pairs to zero with all of $U^\perp$, it also pairs to zero with U (as $u \in U^\perp$), hence with all of $\mathbb{V}$, forcing $u = 0$; so $\omega|_{U^\perp}$ is nondegenerate too.
By induction on n (trivial for $n = 1$, where $U = \mathbb{V}$), a symplectic basis e_2, $\ldots, f_n$ of $U^\perp$ exists; setting $e_1 := v$, $f_1 := w$ gives a symplectic basis of $\mathbb{V} = U \oplus U^\perp$, since cross-pairings between U and $U^\perp$ vanish by definition of $U^\perp$.
Consequently: given nonzero $v, e \in \mathbb{V}$, we can complete both vectors to a symplectic basis as above. We then define the linear map g to send e to v, and each subsequent e-basis vector to the corresponding v-basis vector, thus preserving all pairwise values of ω on basis vectors. Through the bilinearity of ω, g then preserves ω on all of $\mathbb{V}$; and $g(e) = v$. □

Let g be a k-linear automorphism of $\mathbb{V}$ that preserves ω. Since $\mathbb{V}$ is identified with the span of the generators of $A_n(k)$, g assigns to each generator u a linear combination $g(u)$ of generators. Because $A_n(k)$ is presented by generators and the Weyl relations $[v, u] = \omega(u, v)$, and g preserves ω, the elements $g(u)$ satisfy the same relations as the u. Hence the substitution $u \mapsto g(u)$ extends to a unique k-algebra endomorphism Φ_g of $A_n(k)$. Because g is an automorphism, so is Φ_g, and we have $\Phi_g^{-1} = \Phi_{g^{-1}}$.

Proposition 3.2. *Let $0 \neq d \in A_n(k)$ with $N := \deg_B d \geq 1$. There exists an automorphism Φ_g of $A_n(k)$ and $c \in k^\times$ such that*

$$\tilde{d} := c^{-1}\Phi_g(d) = p^N + \sum_{j<N} p^j a_j, \qquad a_j \in C, \quad \deg_B a_j \leq N - j \tag{3.1}$$

in the notation of Convention 2.1. $\tilde{d}$ itself has $\deg_B \tilde{d} = N$.

Proof. Let $s \in k[x, \xi]$ be the Bernstein principal symbol of d, a nonzero homogeneous polynomial of degree N on $\mathbb{V}$. A characteristic-zero field is

infinite, so every nonzero polynomial has a nonvanishing value: we choose $v \in \mathbb{V}$ with $s(v) \neq 0$. Because s is homogeneous of degree $N \geq 1$, $s(0)$ must vanish, therefore v is nonzero. Let $e_p \in \mathbb{V}$ be the standard basis vector corresponding to p. By Lemma 3.1 there is an ω-preserving automorphism h of $\mathbb{V}$ with $h(e_p) = v$. If h preserves ω, then $h^{\mathsf{T}} J h = J$, where J is the Gram matrix of ω. In particular h is invertible, and inverting gives $h^{-1}J^{-1}(h^{\mathsf{T}})^{-1} = J^{-1}$, i.e. $J^{-1} = hJ^{-1}h^{\mathsf{T}}$. Since $J^2 = -\mathrm{Id}$, i.e. $J^{-1} = -J$, this is $hJh^{\mathsf{T}} = J$, so $g := h^{\mathsf{T}}$ again preserves ω. Let Φ_g be the corresponding automorphism of $A_n(k)$, extending $u \mapsto g(u)$ on generators. Let M be the matrix whose rows are the images of the generators, i.e. $g(u_i) = \sum_j M_{ij} u_j$. Since Φ_g sends each generator to a degree-one element, it maps B_m into B_m for all m, and since Bernstein symbols are multiplicative, a standard monomial of degree N is mapped to an element of B_N whose class modulo B_{N-1} is obtained by substituting $u_i \mapsto \sum_j M_{ij} u_j$ in the corresponding commutative monomial. Hence $\Phi_g(d) \equiv s(Mu) \bmod B_{N-1}$. As M is invertible, $s(Mu)$ is again a nonzero homogeneous polynomial of degree N, so $\deg_B \Phi_g(d) = N$ and the Bernstein symbol of $\Phi_g(d)$ is the function $s_g \colon a \mapsto s(Ma)$. Because $g = h^{\mathsf{T}}$, the rows of M are the columns of h, i.e. $Me_j = h(e_j)$. Hence the value of s_g at e_p is

$$s_g(e_p) = s(Me_p) = s(h(e_p)) = s(v) \neq 0.$$

Set $c := s(v)$. Since c is a nonzero scalar, it is a central unit of $A_n(k)$, and $\tilde{d} := c^{-1}\Phi_g(d)$ satisfies $\deg_B \tilde{d} = N$ and $\sigma_N^B(\tilde{d})(e_p) = 1$. Using the normal form $A_n(k) = \bigoplus_j p^j C$, we can write $\tilde{d} = \sum_{j=0}^{N} p^j a_j$ with $a_j \in C$. The Bernstein filtration is compatible with this decomposition, so $N = \deg_B \tilde{d} = \max_j (j + \deg_B a_j)$, and we have $\deg_B a_j \leq N - j$ for all j; in particular $a_j = 0$ for $j > N$ and $a_N \in C \cap B_0 = k$. The symbol is

$$\sigma_N^B(\tilde{d}) = \sum_{j+\deg_B a_j = N} \sigma_N^B(p)^j \, \sigma_N^B(a_j),$$

For $j < N$ the factors $\sigma_N^B(a_j)$ must have a positive degree and must therefore be non-constant in a coordinate other than p, so they vanish at e_p. Because we defined $\tilde{d}$ to be $c^{-1}\Phi_g(d)$, evaluating at e_p gives $a_N = \sigma_N^B(\tilde{d})(e_p) = c^{-1}\sigma_N^B(\Phi_g(d))(e_p) = c^{-1}s_g(e_p) = c^{-1}s(v) = 1$, and therefore

$$\tilde{d} = p^N + \sum_{j<N} p^j a_j, \qquad a_j \in C, \quad \deg_B a_j \leq N - j.$$

□

Formal comparison: Lean proves monicity of the symbol in p with leading coefficient 1, which is all that is used later, not the displayed form

Remark 3.3. Since Φ_g is a ring automorphism, the identity $1 = \tilde{d}R + x_1^N \tilde{d}S$ would transport back to

$$1 = d\big(c^{-1}\Phi_g^{-1}(R)\big) + F\, d\big(c^{-1}\Phi_g^{-1}(S)\big), \qquad F = \Phi_g^{-1}(x_1)^N.$$

Formal comparison: compare

We will now define the modules

$$\mathcal{I} = \tilde{d}A_n + x_1^N \tilde{d}A_n, \qquad Q = A_n/\mathcal{I}. \tag{3.2}$$

We now use the order filtration on $A_n(k)$ (where coordinates have degree zero and derivatives have degree one) to characterize $P = \sigma_N^{\mathrm{ord}}(\tilde{d})$, the order principal symbol of $\tilde{d}$ and show that P annihilates the graded module $\operatorname{gr} Q$.

Definition 3.4. *We say a polynomial $P = \sum_{i=0}^{N} c_i\, \xi_1^i \in k[\xi_1, \dots, \xi_n]$, written as a polynomial in ξ_1 with coefficients $c_i \in k[\xi_2, \dots, \xi_n]$, is* monic of degree N in ξ_1 *if $c_N = 1$.*

Lemma 3.5. *Let $\tilde{d} = p^N + \sum_{j<N} p^j a_j$ with $a_j \in C$, $\deg_B a_j \le N - j$, as in (3.1), and set $N = \deg_B \tilde{d}$. Then $\tilde{d}$ has order N, and its order principal symbol is a polynomial in ξ alone,*

$$\sigma_N^{\mathrm{ord}}(\tilde{d}) = P(\xi), \qquad P \in k[\xi_1, \dots, \xi_n], \qquad \deg P = N. \tag{3.3}$$

P is monic of degree N in ξ_1 and in particular

$$P(1, 0, \dots, 0) = 1. \tag{3.4}$$

Proof. Since $\deg_B \tilde{d} = N$, every PBW monomial $x^\alpha \partial^\beta$ occurring in $\tilde{d}$ satisfies $|\alpha| + |\beta| \le N$. A monomial contributing to the order-N part has $|\beta| = N$, hence $|\alpha| = 0$: it is a pure ∂^β with scalar coefficient. Thus $\tilde{d}$ has order N, and its order-N part is a k-linear combination of ∂^β with $|\beta| = N$, giving $\sigma_N^{\mathrm{ord}}(\tilde{d}) = P(\xi)$, homogeneous of degree N with coefficients in k.
The monomial $p^N = \partial_1^N$ occurs in $\tilde{d}$ with coefficient 1; every other order-N monomial comes from some $p^j a_j$ with $j < N$ and $a_j \in C$ (hence involving no further factor of $p = \partial_1$), so it carries a positive power of some transverse ∂_i, $i > 1$. Such a monomial contributes only to coefficients c_i with $i < N$, never to c_N; hence c_N receives no contribution besides the coefficient of p^N itself, giving $c_N = 1$ identically, i.e. P is monic of degree N in ξ_1. Evaluating at $(1, 0, \dots, 0)$ then gives $P(1, 0, \dots, 0) = 1$. □

Proposition 3.6. *Equip Q with the order filtration $Q_m := qF_m = \operatorname{im}(F_m \to Q)$ induced by the cyclic generator $q = [1]$, the class of 1 in Q and write $P = \sigma_N^{\mathrm{ord}}(\tilde{d})$. Then*

$$P \in \operatorname{Ann}_{k[x,\xi]} \operatorname{gr} Q. \tag{3.5}$$

4. x_1 as a surjective map

We define $E = x_1 p$ and
The first goal of this section will be to show that for Q defined as in Eq. (3.2), we have $Q = Qx_1$ (Proposition 4.3). We will then use this result, as well as Gabber's theorem (Theorem 2.7) to establish two contradictory length estimations for the graded module $\operatorname{gr} Q/(\operatorname{gr} Q)x_1$ that we have assumed to

be nonzero, concluding that therefore, also $\operatorname{gr} Q = (\operatorname{gr} Q)x_1$ (Corollary 4.11).

The commutator of E defines a grading of A_n: x_1 has Euler degree 1, p has Euler degree -1, and the transverse Weyl algebra has Euler degree zero. Let $A_{\geq 0}$ denote the nonnegative part of A_n in this grading. In PBW form (with x_i on the left of ∂_i), $A_{\geq 0}$ is generated by the transverse algebra, E, and x_1. By the commutation relations of the Weyl algebra, it holds that

$$\begin{aligned} Ex_1 &= x_1(E+1), \qquad & x_1E &= (E-1)x_1, \\ Ep &= p(E-1), \qquad & pE &= (E+1)p \end{aligned} \tag{4.1}$$

Additionally, x_1 commutes with the transverse algebra, so x_1 is normal in $A_{\geq 0}$. This also means that every element of Euler degree ≥ 1 lies in $A_{\geq 0}x_1$.

Formal comparison: Lean uses the subring generated by the transverse algebra, x_1 and x_1p, and one-sided normality $x_1r = r'x_1$, which is all that is used

Lemma 4.1. *For every $m \geq 0$,*

$$x_1^m p^m = \prod_{i=0}^{m-1}(E-i), \qquad p^m x_1^m = \prod_{i=1}^{m}(E+i), \tag{4.2}$$

the factors in each product commuting with one another.

Proof. Since both products in (4.2) are polynomials in the single element E, their factors commute. We induct on m; both identities are trivial for $m = 0$. Assuming the first identity for m,

$$\begin{aligned} x_1^{m+1}p^{m+1} &= x_1\Big(\prod_{i=0}^{m-1}(E-i)\Big)p = \Big(\prod_{i=0}^{m-1}(E-1-i)\Big)x_1p \\ &= \Big(\prod_{i=1}^{m}(E-i)\Big)E = \prod_{i=0}^{m}(E-i), \end{aligned}$$

using $x_1E = (E-1)x_1$ repeatedly and the definition of E. The second identity follows identically from $Ep = p(E-1)$. □

Lemma 4.2. *There exists an element $U \in A_{\geq 0}$ such that*

$$1 + Ux_1 \in \mathcal{I}. \tag{4.3}$$

Proof. From Lemma 4.1 we define

$$p^N x_1^N = \prod_{r=1}^{N}(E+r) =: R_+(E), \qquad x_1^N p^N = \prod_{r=0}^{N-1}(E-r) =: R_-(E).$$

The roots of R_+ and R_- are disjoint (the sets $\{-1,\dots,-N\}$ and $\{0,\dots,N-1\}$ of integers, embedded in the prime field $\mathbb{Q} \subset k$), so $\gcd(R_+, R_-) = 1$ in $\mathbb{Q}[E]$. By the extended Euclidean algorithm, there exist $u, v \in \mathbb{Q}[E] \subset k[E]$ with

$$R_+(E)u(E) + R_-(E)v(E) = p^N x_1^N u(E) + x_1^N p^N v(E) = 1.$$

We write $\tilde{d} = p^N + \Sigma$ with $\Sigma = \sum_{j<N} p^j a_j$. Then

$$(\tilde{d}\, x_1^N)u(E) + (x_1^N \tilde{d})v(E) = 1 + w \in \mathcal{I},$$

where

$$w = \Sigma x_1^N u(E) + x_1^N \Sigma v(E).$$

We can write each a_j in Σ in powers of x_1, with coefficients in the transverse Weyl algebra. A term $p^j x_1^r b$ coming from $p^j a_j$ has $r \geq 0$ and $j < N$, and multiplication by x_1^N gives that term Euler degree at least $N - j > 0$. u and v only depend on E, so they have Euler degree 0. Therefore every term of w has positive Euler degree, and thus $w \in A_{\geq 0} x_1$ - or equivalently, there exists a $U \in A_{\geq 0}$, such that $w = Ux_1$. □

Formal comparison: the ring identity is Lean's; the factorization uses left coefficients $c\, x^a p^j$ but that does not change anything because Euler degree does not depend on the order of x and p

Proposition 4.3. *Right multiplication by x_1 on Q is surjective.*

Proof. Let q be the class of 1 in Q. By Equation (4.3), we have $q = -qUx_1$, so $q \in (qA_{\geq 0})x_1$. We use the normality of x_1 to show

$$qA_{\geq 0} = (qA_{\geq 0})x_1. \tag{4.4}$$

The inclusion $\supseteq$ holds because $x_1 \in A^{\geq 0}$ and $A^{\geq 0}$ is a ring. For $\subseteq$, let $r \in A^{\geq 0}$ and choose r' with $x_1 r = r' x_1$ (possible since x_1 is normal) then $qr = -(qU)x_1 r = -(qUr')x_1 \in (qA^{\geq 0})x_1$ since $Ur' \in A^{\geq 0}$.
$qA_{\geq 0}$ is stable under right multiplication by p: If $m = m'x_1$, then

$$mp = m'(x_1 p) = m'E \in qA_{\geq 0}.$$

It is of course also stable under C, since $C = A_{n-1}(k)[x_1] \subseteq A_{\geq 0}$ because no element of C contains p. Since C and p together generate A_n, we have $qA_{\geq 0} = qA_n = Q$. Equation (4.4) is therefore $Q = Qx_1$. □

Formal comparison: Lean argues element by element instead of through the submodule $qA_{\geq 0}$; hypotheses and conclusion are the same

Definition 4.4 ([37][Tag 00CM])**.** *Let R be a commutative ring and $S \subseteq R$ a multiplicative subset (i.e. $1 \in S$ and S is closed under products). The* localization *$S^{-1}R$ is the ring of fractions a/s ($a \in R$, $s \in S$), where*

$$\frac{a}{s} = \frac{a'}{s'} \iff t(s'a - sa') = 0 \quad \text{for some } t \in S,$$

with the usual addition and multiplication of fractions. For an R-module N, the localization *$S^{-1}N$ is the $S^{-1}R$-module of fractions y/s ($y \in N$, $s \in S$), with $y/s = y'/s'$ iff $t(s'y - sy') = 0$ for some $t \in S$. For a prime $\mathfrak{p} \subset R$ we write $R_\mathfrak{p} = S^{-1}R$ and $N_\mathfrak{p} = S^{-1}N$ for $S = R \setminus \mathfrak{p}$. Localization is exact: it sends exact sequences of R-modules to exact sequences of $S^{-1}R$-modules.*

Definition 4.5 ([37][Definition 10.40.1]). *Let R be a ring and let M be an R-module. The* support *of M is the set*

$$\mathrm{Supp}_R(M) = \{\mathfrak{p} \in \mathrm{Spec}(R) | M_{\mathfrak{p}} \neq 0\}.$$

Definition 4.6 ([14, 13], [37, Tags 02LY, 00J3, 00IV]). *The length ℓ_T of a T-module of finite length N is defined as the number of steps in a composition series of N such that*

$$0 = N_0 \subsetneq N_1 \subsetneq \cdots \subsetneq N_{\ell_T} = N$$

where each successive quotient N_i/N_{i-1} is simple (i.e. has no proper nontrivial submodules). The length is well-defined, as it does not depend on the choice of the composition series. Lengths are additive for short exact sequences.

Lemma 4.7. *Let R be a commutative ring, N an R-module and $\mathfrak{p} \subset R$ a prime. Then $\mathfrak{p} \in \mathrm{Supp}_R(N)$ if and only if there is $y \in N$ with $\mathrm{Ann}_R(y) \subseteq \mathfrak{p}$.*

[]

Proof. In $N_{\mathfrak{p}}$, a fraction is zero when it can be cleared by an element outside $\mathfrak{p}$. For $y/1$ this means: there is $u \in R \setminus \mathfrak{p}$ with $uy = 0$. Such a u lies in $\mathrm{Ann}_R(y)$ and not in $\mathfrak{p}$. So

$$y/1 = 0 \iff \mathrm{Ann}_R(y) \not\subseteq \mathfrak{p},$$

and, taking the contrapositive,

$$y/1 \neq 0 \iff \mathrm{Ann}_R(y) \subseteq \mathfrak{p}.$$

($\Leftarrow$) Suppose $y \in N$ has $\mathrm{Ann}_R(y) \subseteq \mathfrak{p}$. Then, $y/1 \neq 0$ in $N_{\mathfrak{p}}$. So $N_{\mathfrak{p}}$ has a nonzero element, hence $N_{\mathfrak{p}} \neq 0$ and $\mathfrak{p} \in \mathrm{Supp}_R(N)$.
($\Rightarrow$) Suppose $N_{\mathfrak{p}} \neq 0$. Pick a nonzero element of $N_{\mathfrak{p}}$. It has the form y/s with $y \in N$ and $s \notin \mathfrak{p}$. The element s is invertible in $N_{\mathfrak{p}}$, so

$$y/s = 0 \iff y/1 = 0.$$

Since $y/s \neq 0$, we get $y/1 \neq 0$, and therefore $\mathrm{Ann}_R(y) \subseteq \mathfrak{p}$, which is the desired y. □

For this part, we write $M = \mathrm{gr}\, Q$ and denote the right multiplication by x_1 by $f : M \to M$. We furthermore define

$$V = \ker(f), \qquad U = \mathrm{coker}(f) = M/x_1 M$$
$$T = k[x_2, \ldots, x_n, \xi_2, \ldots, \xi_n],$$

hence $T[x_1, \xi_1] = k[x, \xi]$. If $n = 1$, we define $T = k$. We want to prove that $U = 0$ by comparing two lengths over a localization of T.

Lemma 4.8. *The modules U and V are finite over T, and $\mathrm{Supp}_T V \subseteq \mathrm{Supp}_T U$. If $\mathfrak{q}$ is a minimal prime in $\mathrm{Supp}_T U$, then $U_{\mathfrak{q}}$ and $V_{\mathfrak{q}}$ have finite length over $T_{\mathfrak{q}}$.*

Proof. The good quotient filtration makes $M = \operatorname{gr} Q$ finitely generated over $\operatorname{gr} A_n = k[x,\xi] = T[x_1,\xi_1]$. Since P annihilates M (Proposition 3.6), the action of $T[x_1,\xi_1]$ on M factors through $S := T[x_1,\xi_1]/(P)$. As P is monic of degree N in ξ_1, in S we have $\xi_1^N = -\sum_{i<N} c_i \xi_1^i$ with $c_i \in T$, so S is generated as a $T[x_1]$-module by $1, \xi_1, \dots, \xi_1^{N-1}$. If $m_1, \dots, m_r$ generate M over S, then the products $\xi_1^i m_s$ $(0 \le i < N,\ 1 \le s \le r)$ generate M over $T[x_1]$. Hence M is finite over $T[x_1]$. Because M is finite, and $T[x_1]$ is a Noetherian ring, the kernel V and cokernel U of multiplication in M by x_1 are also finite over $T[x_1]$. Since they are annihilated by x_1, they are finite over T.

Let $\mathfrak{p} \in \operatorname{Supp}_T(V)$. By Lemma 4.7, we can choose $v \in V = \ker f$ with $\operatorname{Ann}_T(v) \subseteq \mathfrak{p}$. Let us now consider the annihilators of v in $k[x,\xi] = T[x_1,\xi_1]$, defining S as the image of $T \setminus \mathfrak{p}$ in $k[x,\xi]$: $\operatorname{Ann}_{k[x,\xi]}(v) \cap S = \varnothing$, because were $r \in T \setminus \mathfrak{p}$ and in $\operatorname{Ann}_{k[x,\xi]}(v)$, then v would be annihilated by r, which in turn would imply that r is also in $\operatorname{Ann}_T(v)$. Since $\operatorname{Ann}_T(v) \subseteq \mathfrak{p}$, but we assumed $r \in T \setminus \mathfrak{p}$, this is a contradiction. There must therefore exist a prime $\mathfrak{q} \in k[x, \xi]$ with $\mathfrak{q} \cap S = \varnothing$ and $\operatorname{Ann}_{k[x,\xi]}(v) \subseteq \mathfrak{q}$. Since $v \in V = \ker f$, by Lemma 4.7, it follows that $\mathfrak{q} \in \operatorname{Supp}_{k[x,\xi]}(\ker f)$. From Vasconcelos' theorem [22, Theorem 2.4] follows that $\operatorname{Supp}_{k[x,\xi]}(\ker f) \subseteq \operatorname{Supp}_{k[x,\xi]}(\operatorname{coker} f)$: We prove the contrapositive. Let $\mathfrak{q}$ be a prime of $k[x,\xi]$ with $\mathfrak{q} \notin \operatorname{Supp}_{k[x,\xi]}(\operatorname{coker} f)$, i.e. $(\operatorname{coker} f)_\mathfrak{q} = 0$. Since localization is exact, $(\operatorname{coker} f)_\mathfrak{q} = \operatorname{coker}(f_\mathfrak{q})$, so $\operatorname{coker}(f_\mathfrak{q}) = 0$: the endomorphism $f_\mathfrak{q} \colon M_\mathfrak{q} \to M_\mathfrak{q}$ is therefore surjective. The localization $M_\mathfrak{q}$ is a finite module over the Noetherian ring $k[x,\xi]_\mathfrak{q}$, and $f_\mathfrak{q}$ is $k[x,\xi]_\mathfrak{q}$-linear, because f is multiplication by $x_1 \in k[x,\xi]$ and $k[x,\xi]$ is commutative. By Vasconcelos' theorem, a surjective endomorphism of a finite module is injective. Hence $f_\mathfrak{q}$ is injective, and by exactness of localization, $(\ker f)_\mathfrak{q} = \ker(f_\mathfrak{q}) = 0$, i.e. $\mathfrak{q} \notin \operatorname{Supp}_{k[x,\xi]}(\ker f)$. The contrapositive is therefore $\mathfrak{q} \in \operatorname{Supp}_{k[x,\xi]}(\ker f)$, then $\mathfrak{q} \in \operatorname{Supp}_{k[x,\xi]}(\operatorname{coker} f) = \operatorname{Supp}_{k[x,\xi]}(U)$. By Lemma 4.7 there therefore exists an $u \in U$ with $\operatorname{Ann}_{k[x,\xi]}(u) \subseteq \mathfrak{q}$. To get back to the support over T, we define $r \in \operatorname{Ann}_T(u)$. Then $ru = 0$, so $r \in \operatorname{Ann}_{k[x,\xi]}(u) \subseteq \mathfrak{q}$. If $r \notin \mathfrak{p}$, the image of r would lie in $\mathfrak{q} \cap S = \varnothing$, which is impossible. Hence $r \in \mathfrak{p}$, i.e. $\operatorname{Ann}_T(u) \subseteq \mathfrak{p}$, and Lemma 4.7 gives $\mathfrak{p} \in \operatorname{Supp}_T(U)$. Since $\mathfrak{p} \in \operatorname{Supp}_T(V)$ was arbitrary, $\operatorname{Supp}_T(V) \subseteq \operatorname{Supp}_T(U)$. □

Lemma 4.9. *For every minimal prime* $\mathfrak{q}$ *in* $\operatorname{Supp}_T U$,

$$\ell_{T_\mathfrak{q}}(U_\mathfrak{q}) \le \ell_{T_\mathfrak{q}}(V_\mathfrak{q}).$$

Proof. To prove the inequality we set up a spectral sequence measuring how $\ker(f)$ and $\operatorname{coker}(f)$ acting on Q itself compare, order by order, to V and U, acting on $\operatorname{gr} Q$.

Consider the filtered complex $Q \xrightarrow{f} Q$, with decreasing filtration $G^p = F_{-p}Q$, where F is the order filtration. Since x_1 has order 0, $f(G^p) \subseteq G^p$ for every p; the induced map on G^p/G^{p+1}, summed over p, is exactly the action of x_1 on $M = \operatorname{gr} Q$. This is the crudest approximation to $\ker_Q(f)$ and $\operatorname{coker}_Q(f)$: it

correctly identifies V, U as approximations, but it does not yet see elements that vanish to higher order than this naive count predicts. The spectral sequence refines this approximation in successive stages, called *pages*: for each $r \geq 0$, the source and target pages are

$$A_r = \bigoplus_{p\in\mathbb{Z}} \frac{G^p \cap f^{-1}(G^{p+r})}{G^{p+1} \cap f^{-1}(G^{p+r})}, \qquad B_r = \bigoplus_{p\in\mathbb{Z}} \frac{G^p}{(G^p \cap f(G^{p-r+1})) + G^{p+1}},$$

where the degree-p summand of A_r isolates elements of G^p whose image under f drops r extra filtration levels (candidates for the kernel, accurate to order r), and the degree-p summand of B_r isolates the part of G^p not yet hit by f from $r-1$ levels shallower.

Applying f to a representative gives a well-defined map $d_r : A_r \to B_r$, shifting the summand index by r; one checks directly from these definitions that

$$A_0 = B_0 = M, \qquad d_0 = x_1, \qquad A_{r+1} = \ker d_r, \qquad B_{r+1} = \operatorname{coker} d_r,$$

so that each page is obtained from the previous one by taking kernel and cokernel of d_r, exactly as in the construction of a spectral sequence from a filtered complex.

These pages carry a T-module structure. The pages A_r, B_r are subquotients of Q, which a priori carries an action of the transverse Weyl algebra $A_{n-1}(k)$ by right multiplication. This action commutes with f, since the transverse algebra commutes with x_1, so the maps d_r are compatible with it. Two transverse elements commute only up to a commutator, which lowers the order filtration by one. On the page quotients such lower-order terms vanish, so the induced actions commute, and the commutative ring T acts on every A_r and B_r, via the symbols of the transverse generators. The maps d_r and the successor-page maps are T-linear. In particular, this means that localizing at $T \setminus \mathfrak{q}$ and measuring lengths over $T_\mathfrak{q}$ is a well-defined action for the pages.

Now we fix a minimal prime $\mathfrak{q}$ in $\operatorname{Supp}_T U$ and localize at $T\setminus\mathfrak{q}$. We localize the T-modules A_r, B_r and their T-linear maps. Exactness of localization preserves the successor-page kernel and cokernel identities. By Lemma 4.8, $(A_1)_\mathfrak{q} = V_\mathfrak{q}$ and $(B_1)_\mathfrak{q} = U_\mathfrak{q}$ have finite length over $T_\mathfrak{q}$; since each $(A_{r+1})_\mathfrak{q} \subseteq (A_r)_\mathfrak{q}$ and $(B_{r+1})_\mathfrak{q}$ is a quotient of $(B_r)_\mathfrak{q}$, all later pages have finite length as well.

The maps $(B_1)_\mathfrak{q} \to (B_r)_\mathfrak{q}$ are surjective by construction. Let $K_r := \ker((B_1)_\mathfrak{q} \to (B_r)_\mathfrak{q})$. Since $(B_1)_\mathfrak{q} \to (B_{r+1})_\mathfrak{q}$ factors through $(B_1)_\mathfrak{q} \to (B_r)_\mathfrak{q} \to (B_{r+1})_\mathfrak{q}$, we have $K_r \subseteq K_{r+1}$, so $(K_r)_{r\geq 1}$ is an increasing chain of submodules of $(B_1)_\mathfrak{q}$. This chain exhausts $(B_1)_\mathfrak{q}$: Before localization, take a homogeneous class in B_1, represented by $z \in G^p$. Since $Q = Qx_1$ by Proposition 4.3, choose one preimage $z = wx_1$. Exhaustiveness of the filtration means that w must be in G^s for some s; choose $r \geq 1$ with $p - r + 1 \leq s$. Then $w \in G^{p-r+1}$, so z belongs to the denominator of B_r and its class vanishes. Every class is a finite sum of homogeneous classes and hence vanishes on a sufficiently late page. The same holds for a localized class z/a, since its numerator already vanishes on such a page. As $z \in B_1$ was arbitrary, so is z/a in $(B_1)_\mathfrak{q}$, and $\bigcup_{r\geq 1} K_r = (B_1)_\mathfrak{q}$. Since $(B_1)_\mathfrak{q}$ has

finite length, it is a Noetherian $T_{\mathfrak{q}}$-module. By the ascending chain condition, the increasing chain $K_1 \subseteq K_2 \subseteq \cdots$ stabilizes: there exists a finite R with $K_R = K_{R+i}$ for all $i \geq 0$. Combined with $\bigcup_r K_r = (B_1)_{\mathfrak{q}}$, stabilization forces $K_R = (B_1)_{\mathfrak{q}}$. Finally, $K_R = (B_1)_{\mathfrak{q}}$ means $(B_1)_{\mathfrak{q}} \to (B_R)_{\mathfrak{q}}$ must be the zero map: by the first isomorphism theorem, $(B_1)_{\mathfrak{q}}/K_R \cong \operatorname{im}((B_1)_{\mathfrak{q}} \to (B_R)_{\mathfrak{q}})$, and $K_R = (B_1)_{\mathfrak{q}}$ gives $(B_1)_{\mathfrak{q}}/K_R = 0$, so $\operatorname{im}((B_1)_{\mathfrak{q}} \to (B_R)_{\mathfrak{q}}) = 0$. But $(B_1)_{\mathfrak{q}} \to (B_R)_{\mathfrak{q}}$ is also surjective, so $(B_R)_{\mathfrak{q}} = 0$.

At each stage $r \geq 1$ the differential $(d_r)_{\mathfrak{q}}$ gives an exact sequence

$$0 \to (A_{r+1})_{\mathfrak{q}} \to (A_r)_{\mathfrak{q}} \xrightarrow{(d_r)_{\mathfrak{q}}} (B_r)_{\mathfrak{q}} \to (B_{r+1})_{\mathfrak{q}} \to 0,$$

which splits at $\operatorname{im}((d_r)_{\mathfrak{q}})$ into two short exact sequences. Additivity of length gives $\ell((A_r)_{\mathfrak{q}}) = \ell((A_{r+1})_{\mathfrak{q}}) + \ell(\operatorname{im}(d_r)_{\mathfrak{q}})$ and $\ell((B_r)_{\mathfrak{q}}) = \ell(\operatorname{im}(d_r)_{\mathfrak{q}}) + \ell((B_{r+1})_{\mathfrak{q}})$; eliminating $\ell(\operatorname{im}(d_r)_{\mathfrak{q}})$ yields, for every $r \geq 1$,

$$\ell((B_r)_{\mathfrak{q}}) + \ell((A_{r+1})_{\mathfrak{q}}) = \ell((A_r)_{\mathfrak{q}}) + \ell((B_{r+1})_{\mathfrak{q}}). \tag{4.5}$$

Summing this identity over $r = 1, \ldots, R-1$ and reindexing the sums, most terms cancel on both sides, leaving

$$\ell((B_1)_{\mathfrak{q}}) + \ell((A_R)_{\mathfrak{q}}) = \ell((A_1)_{\mathfrak{q}}) + \ell((B_R)_{\mathfrak{q}}).$$

Since $(B_R)_{\mathfrak{q}} = 0$ and $(A_1)_{\mathfrak{q}} = V_{\mathfrak{q}}$, $(B_1)_{\mathfrak{q}} = U_{\mathfrak{q}}$, this reads

$$\ell(U_{\mathfrak{q}}) + \ell((A_R)_{\mathfrak{q}}) = \ell(V_{\mathfrak{q}}).$$

As $\ell((A_R)_{\mathfrak{q}}) \geq 0$, we conclude

$$\ell_{T_{\mathfrak{q}}}(U_{\mathfrak{q}}) \leq \ell_{T_{\mathfrak{q}}}(V_{\mathfrak{q}}).$$

□

[Readable correspondence note.]
On the other hand, we can also establish a different inequality:

Lemma 4.10. *For every minimal prime* $\mathfrak{q}$ *in* $\operatorname{Supp}_T U$*,*

$$\ell_{T_{\mathfrak{q}}}(U_{\mathfrak{q}}) > \ell_{T_{\mathfrak{q}}}(V_{\mathfrak{q}}).$$

Proof. We start by showing, that by Gabber's involutivity theorem, x_1 cannot lie in any minimal prime. We then define a module K as the kernel of multiplication by a power of x_1 over the localized $M_{\mathfrak{q}}$, which allows us to split the length of $U_{\mathfrak{q}}$ into two parts, $\ell(U_{\mathfrak{q}}) = \ell(V_{\mathfrak{q}}) + \ell\big((M_{\mathfrak{q}}/K)/(x_1(M_{\mathfrak{q}}/K))\big)$. It remains to show that the second term is nonzero. For this we use the fact that x_1 does not lie in any minimal prime to deduce that x_1 must therefore be invertible at the minimal primes while lying in every prime of the support of $U_{\mathfrak{q}}$. Nakayama's lemma then shows that $(M_{\mathfrak{q}}/K)/x_1(M_{\mathfrak{q}}/K) \neq 0$.

Let us consider $\operatorname{Ann}_C M$. By Gabber's Theorem (2.7, [16]), $J = \sqrt{\operatorname{Ann}_C M}$ is involutive. If $\mathfrak{p}$ is a minimal prime of J, then $\mathfrak{p}$ is itself involutive, i.e. $\{\mathfrak{p}, \mathfrak{p}\} \subseteq \mathfrak{p}$. We can show this the following way: We localize $k[x, \xi]$ at $\mathfrak{p}$. Since $\mathfrak{p}$

is minimal over J, the only prime of $k[x,\xi]_{\mathfrak{p}}$ containing $J_{\mathfrak{p}}$ is $\mathfrak{p}_{\mathfrak{p}}$ itself; as J is already radical, so is its localization, giving

$$J_{\mathfrak{p}} = \sqrt{J_{\mathfrak{p}}} = \mathfrak{p}_{\mathfrak{p}}.$$

The Poisson bracket extends to $k[x,\xi]_{\mathfrak{p}}$ since by the Leibniz rule, $\{a/s,\, b/t\}$ is determined by $\{a,b\}$ and the brackets of a,b with s,t, giving a well-defined bracket on $k[x,\xi]_{\mathfrak{p}}$ agreeing with the one on $k[x,\xi]$ under $f \mapsto f/1$; in particular $\{f/1, g/1\} = \{f,g\}/1$ for $f,g \in k[x,\xi]$, and the involutivity of J localizes to $\{J_{\mathfrak{p}}, J_{\mathfrak{p}}\} \subseteq J_{\mathfrak{p}}$. Now take $f,g \in \mathfrak{p}$. Then $f/1, g/1 \in \mathfrak{p}_{\mathfrak{p}} = I_{\mathfrak{p}}$, so

$$\{f,g\}/1 = \{f/1, g/1\} \in J_{\mathfrak{p}} = \mathfrak{p}_{\mathfrak{p}}.$$

Thus there is $t \notin \mathfrak{p}$ with $t\{f,g\} \in \mathfrak{p}$. Since $\mathfrak{p}$ is prime and $t \notin \mathfrak{p}$, this forces $\{f,g\} \in \mathfrak{p}$.

From the involutivity of $\mathfrak{p}$ follows that no $\mathfrak{p}$ contains x_1: Assume $x_1 \in \mathfrak{p}$. Since $P \in \mathfrak{p}$, also the Poisson brackets

$$\{x_1, P\}, \quad \{x_1, \{x_1, P\}\}, \dots$$

lie in $\mathfrak{p}$. $\{x_1, -\}$ differentiates once with respect to ξ_1, and because P is homogeneous of degree N and monic in ξ_1, its Nth derivative is the nonzero constant $N!$ - therefore the Nth bracket would put $N!$ in $\mathfrak{p}$. Since $\mathfrak{p}$ is a proper prime ideal, no nonzero constant can be in $\mathfrak{p}$, so we must conclude $x_1 \notin \mathfrak{p}$. This result holds under localization at $T \setminus \mathfrak{q}$: no minimal prime of $\mathrm{Ann}_{k[x,\xi]_{\mathfrak{p}}}(M_{\mathfrak{q}})$ contains x_1 either, since the annihilator of the finite module M commutes with localization.

Consider the increasing chain of kernels of powers of the localization $f_{\mathfrak{q}}$ of right multiplication with x_1 $\ker(f_{\mathfrak{q}}) \subseteq \ker(f_{\mathfrak{q}}^2) \subseteq \dots$. Since $M_{\mathfrak{q}}$ is Noetherian, this chain stabilizes. Let $K := \ker(f_{\mathfrak{q}}^m)$ at a stage where it stabilizes, so $f_{\mathfrak{q}}(K) = x_1 K \subseteq K$ and $\ker(f_{\mathfrak{q}}^{m+1}) = \ker(f_{\mathfrak{q}}^m) = K$. Each quotient $\ker(f_{\mathfrak{q}}^{i+1})/\ker(f_{\mathfrak{q}}^i)$ maps to $V_{\mathfrak{q}} = \ker(f_{\mathfrak{q}})$ injectively via $v \mapsto {x_1}^i v$, so, as $V_{\mathfrak{q}}$ has finite length, so does each quotient and hence K itself.

Two elementary facts about K:

- ${f_{\mathfrak{q}}}^{-1}K = K$: if $v \in K$ then $f_{\mathfrak{q}}(v) \in K$ (as $f_{\mathfrak{q}}(K) \subseteq K$); if $f_{\mathfrak{q}}(v) \in K = \ker(f_{\mathfrak{q}}^m)$ then $f_{\mathfrak{q}}^{m+1}(v) = 0$, so $v \in \ker({f_{\mathfrak{q}}}^{m+1}) = K$.
- $K \cap x_1 M_{\mathfrak{q}} = x_1 K = f_{\mathfrak{q}}(K)$: "$\supseteq$" is trivial. For "$\subseteq$", if $w = f_{\mathfrak{q}}(v) \in K$ for some $v \in M_{\mathfrak{q}}$, then $v \in f_{\mathfrak{q}}^{-1}K = K$, so $w \in f_{\mathfrak{q}}(K) = x_1 K$.

For any endomorphism φ of a finite-length module N, additivity along $0 \to \ker\varphi \to N \to \varphi(N) \to 0$ and $0 \to \varphi(N) \to N \to N/\varphi(N) \to 0$ gives $\ell(\ker\varphi) = \ell(N/\varphi(N))$. We can apply this to $\varphi = f_{\mathfrak{q}}|_K$: since $\ker(f_{\mathfrak{q}}) \subseteq K$, we have $\ker(f_{\mathfrak{q}}|_K) = \ker(f_{\mathfrak{q}}) = V_{\mathfrak{q}}$, so with $f_{\mathfrak{q}}(K) = x_1 K$,

$$\ell(K/x_1K) = \ell(V_{\mathfrak{q}}).$$

On the other hand, the natural surjection $M_{\mathfrak{q}}/(x_1 M_{\mathfrak{q}}) \to (M_{\mathfrak{q}}/K)/(x_1(M_{\mathfrak{q}}/K))$ has the kernel $(K + (x_1 M_{\mathfrak{q}}))/(x_1 M_{\mathfrak{q}}) \cong K/(K \cap (x_1 M_{\mathfrak{q}})) = K/(x_1 K)$, giving (we now write $E = (M_{\mathfrak{q}}/K)$ for better readability) the exact sequence

$$0 \longrightarrow K/(x_1K) \longrightarrow U_{\mathfrak{q}} \longrightarrow E/(x_1E) \longrightarrow 0,$$

so by additivity and using $\ell(K/(x_1K)) = \ell(V_\mathfrak{q})$,

$$\ell(U_\mathfrak{q}) = \ell(V_\mathfrak{q}) + \ell\big(E/(x_1E)\big).$$

It remains to show the last term is positive.
Since $\mathfrak{q} \in \operatorname{Supp}_T U$, $U_\mathfrak{q} = M_\mathfrak{q}/(x_1M_\mathfrak{q}) \neq 0$. We choose $\mathfrak{P} \in \operatorname{Supp}_{k[x,\xi]_\mathfrak{p}}(M_\mathfrak{q}/(x_1M_\mathfrak{q}))$ and a minimal prime $\mathfrak{p} \subseteq \mathfrak{P}$ of $M_\mathfrak{q}$. We have shown above that $x_1 \notin \mathfrak{p}$, so x_1 is a unit in the localization at $\mathfrak{p}$; as K is annihilated by a power of x_1, $K_\mathfrak{p} = 0$. From $0 \to K_\mathfrak{p} \to (M_\mathfrak{q})_\mathfrak{p} \to E_\mathfrak{p} \to 0$ we get $E_\mathfrak{p} \cong (M_\mathfrak{q})_\mathfrak{p} \neq 0$, so $\mathfrak{p}$ lies in the support of $M_\mathfrak{q}/K$ in $k[x,\xi]_\mathfrak{p}$. Since support is closed under passing to larger primes [37, Lemma 10.40.5 (Tag 00L2)] and $\mathfrak{p} \subseteq \mathfrak{P}$, also $\mathfrak{P} \in \operatorname{Supp}_{k[x,\xi]_\mathfrak{p}} E$, i.e. $E_\mathfrak{P} \neq 0$.
Since $\mathfrak{P} \in \operatorname{Supp}_{k[x,\xi]_\mathfrak{p}}(M_\mathfrak{q}/(x_1M_\mathfrak{q}))$ and $M_\mathfrak{q}/(x_1M_\mathfrak{q})$ is annihilated by x_1, we have $x_1 \in \mathfrak{P}$. If $(x_1E_\mathfrak{P}) = E_\mathfrak{P}$, then as x_1 lies in the maximal ideal of the local ring $(k[x,\xi]_\mathfrak{p})_\mathfrak{P}$, Nakayama's lemma [23], (see also [37, Tag 00DV]) would force $E_\mathfrak{P} = 0$, a contradiction. Hence $\big(E/(x_1E)\big)_\mathfrak{P} \neq 0$, so $E/(x_1E) \neq 0$ and $\ell\left(E/(x_1E)\right) > 0$.
Therefore $\ell(U_\mathfrak{q}) = \ell(V_\mathfrak{q}) + \ell\big(E/x_1E\big) > \ell(V_\mathfrak{q})$. □

Corollary 4.11. *Multiplication by x_1 on $\operatorname{gr} Q$ is surjective, i.e. $\operatorname{gr} Q = (\operatorname{gr} Q)\, x_1$.*

Proof. We write $M = \operatorname{gr} Q$ and $U = M/x_1M$. If $U \neq 0$, it has a minimal support prime $\mathfrak{q}$. Lemma 4.8 makes the relevant lengths finite, and Lemmas 4.9 and 4.10 contradict one another. Thus $U = M/x_1M = 0$ and therefore $\operatorname{gr} Q = (\operatorname{gr} Q)\, x_1$. □

5. Avoiding the hyperplane $\{x_1 = 0\}$

This section uses mainly vocabulary and concepts from algebraic geometry. We will define the most important quantities used here (for a comprehensive overview of the concepts of algebraic geometry, we refer to textbooks such as [**?**]), then state the two main results. Theorem 5.5 shows that for a closed irreducible subvariety $Y \subseteq X$ not meeting the hyperplane $H = \{x_1 = 0\}$, the covector dx_1 is an asymptotic conormal direction of Y. Theorem 5.6 uses it to show that no nonempty closed, fibre-conical, involutive subset $W \subseteq T^*X$ that lies over the complement of H can be contained in the zero set of a polynomial $P(\xi)$ with $P(1,0,\dots,0) \neq 0$. We prove the first theorem in Subsection 5.2, with the lemmas stated and proved where they are used, and the second in Subsection 5.3.

5.1. **Definitions.** Throughout this section K denotes an algebraically closed field of characteristic zero, and $X = K^n$ is affine n-space with *affine coordinates* $x = (x_1, \dots, x_n)$. Since X has a global coordinate frame, $T^*X = X \times (K^n)^*$, so all cotangent fibres are identified with $(K^n)^*$. We write $\pi\colon T^*X \to X$ for the projection forgetting the covector.

The projective space $\mathbb{P}^n$ has *homogeneous coordinates* $[Z_0 : \cdots : Z_n]$: a nonzero vector $(Z_0, \ldots, Z_n) \in K^{n+1}$, taken up to multiplication by a common nonzero scalar. We embed $X \hookrightarrow \mathbb{P}^n$ by $(x_1, \ldots, x_n) \mapsto [1 : x_1 : \cdots : x_n]$. Its image is the chart $\{Z_0 \neq 0\}$, and a point of this chart with homogeneous coordinates $(Z_0, \ldots, Z_n)$ has affine coordinates $x_i = Z_i/Z_0$ (dehomogenization). The complement $\{Z_0 = 0\}$ is the hyperplane at infinity. A vector in K^{n+1} representing a point of $\mathbb{P}^n$ is called a homogeneous column vector; analogously to the affine coordinates, we call its entry i its x_i-coordinate, entry 0 being the homogenizing coordinate. For a row vector $\ell = (\ell_0, \ldots, \ell_n)$, its *affine part* is $(\ell_1, \ldots, \ell_n)$.

Definition 5.1. *If $K \subseteq L$ are fields, we call L an extension of K and write L/K. For $z \in L$ and $B \subseteq L$ we say:*

(1) z is algebraic over K *if there exists a nonzero polynomial g in one variable over K with $g(z) = 0$;*
(2) z is transcendental over K *if no such polynomial exists, i.e. $g(z) = 0 \Rightarrow g = 0$;*
(3) κ/K is algebraic *if every element of κ is algebraic over K;*
(4)
(5) B is algebraically independent *over K if no nonzero polynomial over K in finitely many elements of B vanishes.*
(6) A maximal algebraically independent subset is a transcendence basis *of L/K. If B is a transcendence basis, then L is algebraic over the subfield $K(B)$ generated by K and B.*
(7) Any two transcendence bases of L/K have the same cardinality, called the transcendence degree $\operatorname{trdeg}_K L$.

The elements of κ that are algebraic over K form a subfield, and every finite set generating κ over K contains a transcendence basis.

Definition 5.2. *Let κ/K be a field extension.*

*(a) A K-*derivation *of κ is a K-linear map $D\colon \kappa \to \kappa$ with $D(ab) = aD(b) + bD(a)$.*
(b) Let $\mathcal{G}$ be the free κ-vector space with basis symbols $\langle a \rangle$, $a \in \kappa$. The space of differentials $\Omega_{\kappa/K}$ *is the quotient of $\mathcal{G}$ by the subspace spanned by*

$$\langle a+b\rangle - \langle a\rangle - \langle b\rangle, \quad \langle ab\rangle - a\langle b\rangle - b\langle a\rangle, \quad \langle c\rangle \qquad (a, b \in \kappa,\ c \in K),$$

and $da \in \Omega_{\kappa/K}$ denotes the class of $\langle a \rangle$.
(c) A K-derivation D induces the κ-linear map $\Omega_{\kappa/K} \to \kappa$, $da \mapsto D(a)$, since the linear map $\langle a \rangle \mapsto D(a)$ on $\mathcal{G}$ kills the relations. Conversely, a κ-linear map $\Omega_{\kappa/K} \to \kappa$ gives the K-derivation $a \mapsto$ (its value on da). Hence a basis $d\beta_1, \ldots, d\beta_d$ of $\Omega_{\kappa/K}$ has dual derivations $D_1, \ldots, D_d$ with $D_k(\beta_l) = \delta_{kl}$.

Definition 5.3. *For Y as above, the set of* conormal directions *is*

$$\operatorname{Dir}(Y) = \overline{\{[\xi] \in \mathbb{P}((K^n)^*) : 0 \neq \xi \in N(Y)\}} \subseteq \mathbb{P}((K^n)^*),$$

the Zariski closure of the projectivization of $N(Y)$.

Definition 5.4. *A closed subset $W \subseteq T^*X$ is* fibre-conical *if it is stable under the fibrewise scaling $(y, \xi) \mapsto (y, t\xi)$, $t \in \mathbb{G}_m$.*

Theorem 5.5. *Let $Y \subseteq X$ be nonempty, closed, and irreducible with $Y \cap H = \varnothing$, and let $N(Y) = \bigcup_{y \in Y_{\mathrm{sm}}} N_y^* Y \subseteq K^n$ be the conormal cone over the smooth locus Y_{sm}.*

(i) Every polynomial $P \in K[\xi_1, \dots, \xi_n]$ vanishing on $N(Y)$ satisfies $P(1, 0, \dots, 0) = 0$.
(ii) $[dx_1] \in \mathrm{Dir}(Y)$.

Formal comparison: the formal proof follows the same route except for a transcendence argument in one sublemma

Theorem 5.6. *Let K be an algebraically closed field of characteristic zero, and let $W \subseteq T^*X$ be a Zariski closed subset that is fibre-conical and involutive. Let $P \in K[\xi_1, \dots, \xi_n]$ be a polynomial with $P(1, 0, \dots, 0) \neq 0$. If P vanishes on W and $W \cap \pi^{-1}(H) = \varnothing$, then W must be empty.*

Formal comparison: any algebraically closed field of characteristic 0; P need not be homogeneous

5.2. **Proof of Theorem 5.5.**

Proof. If Y is nonempty, closed and irreducible, then $\mathfrak{p} := I(Y)$ is a prime ideal and $Y = V(\mathfrak{p})$. We will write $R = K[x]/\mathfrak{p}$, $F = \mathrm{Frac}(R)$ for the field of fractions, and $\bar{x}_i$ for the image of x_i in R.

If $\bar{x}_1$ is algebraic over K, there exists a nonzero polynomial g over K in one variable with $g(\bar{x}_1) = 0$ (Def. 5.1), therefore $g(x_1) \in \mathfrak{p}$. Since K is algebraically closed, $g(x_1) = \lambda \prod_i (x_1 - c_i)$ with $\lambda \in K^\times$, and as $\mathfrak{p}$ is prime, some factor $x_1 - c$ lies in $\mathfrak{p}$. Thus $x_1 = c$ on $Y = V(\mathfrak{p})$, and since $Y \cap \{x_1 = 0\} = \varnothing$ and $Y \neq \varnothing$, we have $c \neq 0$. In this case $f = x_1 - c$ lies in $\mathfrak{p} = I(Y)$ and has the differential $df_y = dx_1$ at every point $y \in Y_{\mathrm{sm}}$, so $dx_1 \in N_y^* Y$, and its coordinate vector in the dual basis $(1, 0, \dots, 0) \in N(Y)$. Every polynomial vanishing on $N(Y)$ therefore vanishes at $(1, 0, \dots, 0)$, and $[dx_1] \in \mathrm{Dir}(Y)$. So we may assume from now on that $\bar{x}_1$ is transcendental over K.

The rest of the proof follows this rough logic. We construct a discrete valuation ring V of F at infinity in which x_1 vanishes, and the corresponding generic point $y \in \kappa((t))^n$ of Y. The tangent cone $\mathcal{C}$ of Y at y has a basis M of power series in t that remains a basis at $t = 0$, and whose reduction $M(0)$ has vanishing x_1-coordinate. Hence the row vector $u = (0, 1, 0, \dots, 0)$, which reads off the x_1-coordinate, satisfies $uM(0) = 0$, and we lift it to row vectors $\ell(t)$ annihilating $\mathcal{C}$, with $\ell(0) = u$. By construction the affine part

$\tilde{\ell}(t)$ consists of conormal covectors to Y at y, and $\tilde{\ell}(0) = dx_1$. A polynomial vanishing on $N(Y)$ therefore vanishes at dx_1, and so $[dx_1] \in \mathrm{Dir}(Y)$.
We set $\bar{x}_0 := 1$.

Lemma 5.7. *Let $\bar{x}_1$ be transcendental over K. There exist a valuation ring W with $K \subseteq W \subset F$, $\bar{x}_1 \in \mathfrak{m}_W$, and an index j with $\bar{x}_i/\bar{x}_j \in W$ for all $0 \le i \le n$.*

Proof. ince $\bar{x}_1$ is transcendental over K, $S = K[\bar{x}_1]$ is a polynomial ring, and $\mathfrak{q} = \bar{x}_1 S$ is a prime ideal of S. The localization $S_{\mathfrak{q}} \subseteq F$ is a local ring with maximal ideal $\bar{x}_1 S_{\mathfrak{q}}$. By [37, Tag 00IA, Lemma 10.50.2] there exists a valuation ring W of F dominating $S_{\mathfrak{q}}$, i.e. $S_{\mathfrak{q}} \subseteq W$ and $\mathfrak{m}_{S_{\mathfrak{q}}} = \mathfrak{m}_W \cap S_{\mathfrak{q}}$. Hence $K \subseteq W$ and $\bar{x}_1 \in \mathfrak{m}_{S_{\mathfrak{q}}} \subseteq \mathfrak{m}_W$. For nonzero $a, b \in F$ either $a/b \in W$ or $b/a \in W$, so among the finitely many nonzero $\bar{x}_i$ we can choose j with $\bar{x}_i/\bar{x}_j \in W$ for all i (note $\bar{x}_0 = 1 \neq 0$). □

Let W be the valuation ring of Lemma 5.7. We define the column vector q in homogeneous coordinates by

$$q_i := \bar{x}_i/\bar{x}_j \qquad (0 \le i \le n),$$

so that $q_i \in W$, $q_j = 1$, $q_0 = \bar{x}_j^{-1} \neq 0$, $\bar{x}_i = q_i/q_0$ and $q_1 = q_0 \bar{x}_1$.

Lemma 5.8. *For q defined as above, there exists a discrete valuation ring V with $K \subseteq V \subset F$, $\operatorname{Frac} V = F$, such that, with $\kappa = V/\mathfrak{m}_V$ and $\operatorname{res}(q_i)$ the residue of q_i in κ:*

(a) $q_i \in V$ for all i, $\bar{x}_1 \in \mathfrak{m}_V$ and $R \not\subseteq V$;
(b) κ is algebraic over $L = K(\operatorname{res}(q_0), \dots, \operatorname{res}(q_n))$;
(c) $\Omega_{V/K}$ is a finitely generated V-module.

Proof. Let $C_0 = K[q_0, \dots, q_n, \bar{x}_1] \subseteq W$; it is finitely generated with $\operatorname{Frac} C_0 = F$ since on one hand $\bar{x}_i = q_i/q_0$ so every $\bar{x}_i$ is in $\operatorname{Frac} C_0$ and on the other hand, F is generated by the $\bar{x}_i$. Let A be its integral closure in F. Since C_0 is a domain finitely generated over a field, A is a finite C_0-module, hence a finitely generated, integrally closed Noetherian domain with $\operatorname{Frac} A = F$ and $q_i \in A$. As W is a valuation ring and thereby is integrally closed, $A \subseteq W$. $\bar{x}_1 \in \mathfrak{m}_W$, so x_1 is a nonzero nonunit of W and therefore also of A. Since $\bar{x}_1$ is a nonzero nonunit of the Noetherian domain A, the ideal $\bar{x}_1 A$ is proper, so some prime $\mathfrak{p}_D$ is minimal among the primes containing $\bar{x}_1$. By Krull's principal ideal theorem [37, Lemma 10.60.11], $\operatorname{ht} \mathfrak{p}_D \le 1$. Because A is a domain, the only prime of height 0 is (0), and $\bar{x}_1 \neq 0$ lies in $\mathfrak{p}_D$; hence $\operatorname{ht} \mathfrak{p}_D = 1$.
Put $V := A_{\mathfrak{p}_D}$. This is a Noetherian local domain. It is normal, since a localization of an integrally closed domain is integrally closed, and its dimension equals $\operatorname{ht} \mathfrak{p}_D = 1$. A Noetherian local normal domain of dimension one is a discrete valuation ring [37, Lemma 10.119.7], so V is a discrete valuation ring.
It remains to check the listed properties.

- $K \subseteq A \subseteq V$: we have $K \subseteq C_0 \subseteq A$, and the domain A embeds into its localization V.
- $\operatorname{Frac} V = F$: since $A \subseteq V \subseteq F$ and $\operatorname{Frac} A = F$.
- $q_i \in V$ for all i: since $q_i \in A \subseteq V$.
- $\bar{x}_1 \in \mathfrak{m}_V$: since $\mathfrak{m}_V = \mathfrak{p}_D A_{\mathfrak{p}_D}$ and $\bar{x}_1 \in \mathfrak{p}_D$. Since $Y \cap H = Y \cap \{x_1 = 0\} = \varnothing$, the ideal $(\mathfrak{p}, x_1)$ has empty zero set, so $1 = p + gx_1$ for some $p \in \mathfrak{p}$, $g \in K[x]$ (by the Nullstellensatz, Theorem 2.8). Modulo $\mathfrak{p}$ this gives $1 = \bar{g}\,\bar{x}_1$, so $\bar{x}_1 \in R^\times$. If $R \subseteq V$, then $\bar{x}_1 \in R^\times \subseteq V^\times$, contradicting $\bar{x}_1 \in \mathfrak{m}_V$.

This proves (a).
(b) Since $V = A_{\mathfrak{p}_D}$ and $\mathfrak{m}_V = \mathfrak{p}_D V$, we have $\kappa = V/\mathfrak{m}_V = (A/\mathfrak{p}_D)_{(0)} = \operatorname{Frac}(A/\mathfrak{p}_D)$, and the residue map $\operatorname{res} : V \to \kappa$ restricts to the quotient map $A \to A/\mathfrak{p}_D$. As A is integral over C_0 and $\bar{x}_1 \in \mathfrak{p}_D$, the ring $A/\mathfrak{p}_D$ is integral over the image of C_0 in $A_{\mathfrak{p}_D}$, which is $K[\operatorname{res}(q_0), \dots, \operatorname{res}(q_n)] \subseteq L$. So every element of $A/\mathfrak{p}_D$, and hence of its fraction field κ, is algebraic over L.
(c) A is a finitely generated K-algebra, so $\Omega_{A/K}$ is a finite A-module, and $\Omega_{V/K} = V \otimes_A \Omega_{A/K}$ since V is a localization of A. □

By Lemma 5.8 there exists a discrete valuation ring $V \subset F$ with $q_i \in V$ for all i, $\bar{x}_1 \in \mathfrak{m}_V$ and $R \not\subseteq V$. Let ϖ be a generator of $\mathfrak{m}_V$, then every nonzero $f \in F$ can be written as $f = \varpi^k \cdot u$ with $k \in \mathbb{Z}$ and u a unit of V - we say $\operatorname{ord}_V(f) = k \in \mathbb{Z}$ is the valuation of f in V. We denote the residual field with $\kappa = V/\mathfrak{m}_V$.
Let $\widehat{V}$ be the completion of V. Since $V \supseteq K$ and K is of characteristic zero, there exists a coefficient field of $\widehat{V}$ containing K and mapping isomorphically onto κ [37, Tag 0C0S]; choosing one gives us an isomorphism $\widehat{V} \cong \kappa[[t]]$ under which ϖ corresponds to the formal variable t. Since the natural map $V \to \widehat{V}$ is injective ($\bigcap_k \varpi^k V = 0$) and extends to an embedding $F \hookrightarrow \kappa((t))$, we can identify elements of F with Laurent series. The residue $\operatorname{res}(f)$ of $f \in V$ is then just the constant term $f(0)$ of its series; in particular $\operatorname{res}(q_i) = q_i(0)$.
Let $a = \operatorname{ord}_V(q_0)$ and $b = \operatorname{ord}_V(q_1)$. $a, b \geq 0$ since q_0, q_1 are elements of V, so that in $\kappa[[t]]$

$$q_0(t) = t^a u_0(t), \qquad q_1(t) = t^b u_1(t),$$

with $u_0(0) \neq 0 \neq u_1(0)$. Since $\bar{x}_1 \in \mathfrak{m}_V$, $b - a = \operatorname{ord}_V(q_1/q_0) = \operatorname{ord}_V(\bar{x}_1) > 0$. Moreover $a > 0$: if $a = 0$, then q_0 is a unit of V, hence all $\bar{x}_i = q_i/q_0$ lie in V, so $R \subseteq V$ since also $K \subseteq V$, contradicting Lemma 5.8(a). It follows that $j \notin \{0, 1\}$ since $q_j = 1$ is a unit while q_0, q_1 are not.
Dividing by $q_0(t) \neq 0$, we define $y = (q_1(t)/q_0(t), \dots, q_n(t)/q_0(t)) \in \kappa((t))^n$ in affine coordinates. its coordinates are the images of $\bar{x}_1, \dots, \bar{x}_n$ under $F \hookrightarrow \kappa((t))$. Since ring homomorphisms commute with polynomial evaluation and this map is injective, for every polynomial $f \in K[x_1, \dots, x_n]$ we have

$$f(y) = 0 \iff f(\bar{x}_1, \dots, \bar{x}_n) = 0 \iff f \in \mathfrak{p}. \tag{5.1}$$

Let $df_y = \nabla f(y) \in \kappa((t))^n$ for a polynomial f. We define Θ_y to be the tangent space of Y at the generic point y,

$$\Theta_y = \bigcap_{f \in \mathfrak{p}} \ker df_y \subseteq \kappa((t))^n,$$

the directions v with $df_y \cdot v = 0$ for all equations f of Y. We now pass to homogeneous coordinates by defining a map

$$\rho\colon \kappa((t))^{n+1} \to \kappa((t))^n, \qquad \rho(v(t))_i = \frac{v_i(t)q_0(t) - q_i(t)v_0(t)}{q_0^2(t)} \quad (1 \le i \le n),$$

which sends a homogeneous vector $v(t)$ to the affine direction it induces at $q(t)$ (by the quotient rule); its kernel is the line $\kappa((t))q(t)$. We write $\mathcal{C} = \rho^{-1}(\Theta_y)$ for the affine cone over the projective tangent plane. Our goal is to build a basis of $\mathcal{C}$ whose entries are power series in t and which also remains a basis at $t = 0$. We do so by building independent vectors from the tangent planes of $q(t)$. Since $0 < a < b$, in κ we have $q_0(t) = q_1(0) = 0$ and $q_j(0) = 1$. Let $L = K(q_0(0), \dots, q_n(0)) \subseteq \kappa$; by Lemma 5.8(b), κ/L is algebraic.

Lemma 5.9. *There exist indices $i_1, \dots, i_d \notin \{0, 1, j\}$ such that $d(q_{i_1}(0)), \dots, d(q_{i_r}(0))$ form a basis of $\Omega_{\kappa/K}$. In particular $d' = \dim_\kappa \Omega_{\kappa/K} < \infty$.*

Proof. Since $q_0(0) = q_1(0) = 0$ and $q_j(0) = 1$, the field L is generated over K by the (finitely many) $q_i(0)$ with $i \notin \{0, 1, j\}$. By Definition 5.1, some subset $B = \{q_{i_1}(0), \dots, q_{i_{d'}}(0)\}$ of these is a transcendence basis of L/K; which defines $d' =$. As $L/K(B)$ and κ/L are algebraic, so is $\kappa/K(B)$.

By Definition 5.2, the differentials da, $a \in \kappa$, span $\Omega_{\kappa/K}$, so it suffices to show $da \in \operatorname{span}_\kappa\{d(q_{i_k}(0))\}$ for an arbitrary $a \in \kappa$. Fix $a \in \kappa$. Since $\kappa/K(B)$ is algebraic, a has a minimal polynomial g over $K(B)$, i.e. the monic polynomial of least degree over $K(B)$ with $g(a) = 0$. Since K has characteristic zero, $g'(a) \neq 0$. Applying d to $g(a) = 0$ and defining $\partial_k := \partial_{q_{i_k}(0)}$ gives $g'(a)\, da + \sum_k (\partial_k g)(a)\, d(q_{i_k}(0)) = 0$, so $da \in \operatorname{span}_\kappa\{d(q_{i_k}(0))\}$. As the da span $\Omega_{\kappa/K}$, so do the $d(q_{i_k}(0))$.

$K(B)$ is a rational function field, so the partial derivatives ∂_k are K-derivations of $K(B)$, and each extends to a K-derivation D_k of the separable algebraic extension κ (by differentiating the minimal polynomials). Then $D_k(q_{i_l}(0)) = \delta_{kl}$, so if $\sum_l c_l\, d(q_{i_l}(0)) = 0$, applying D_k gives $c_k = 0$, i.e. they are linearly independent. □

We write for all $q_i(t) = \sum_r c_{ir} t^r$. The derivative with respect to t is then $q_i'(t) = \sum_r r\, c_r t^{r-1}$. This is a K-derivation of $\kappa((t))$. By Lemma 5.9 there exist $i_1, \dots, i_{d'} \notin \{0, 1, j\}$ such that $\{d(q_{i_1}(0)), \dots, d(q_{i_{d'}}(0))\}$ form a basis of $\Omega_{\kappa/K}$. Let $D_1, \dots, D_{d'}$ be the dual K-derivations of this basis (Definition 5.2(c)). We apply D_k coefficientwise to $q_i(t) = \sum_r c_{ir} t^r$ and define $Z_{ik}(t) := \sum_r D_k(c_{ir}) t^r$. Since $D_k(q_{i_l}(0)) = \delta_{kl}$, the reduction $Z(0)$ is the identity on the rows $i_1, \dots, i_d$, and since $q_j(t) = 1$ is constant, $Z_{jk}(t) = 0$. Rows

0 and 1 of $Z(t)$ have order at least a and b, since D_k does not differentiate with respect to t.
Let $\tilde{Z}(t)$ be the $d' \times d'$ submatrix of $(t)Z$ consisting of the rows $i_1, \dots, i_{d'}$ and all columns. Its reduction $\tilde{Z}(0)$ is the identity matrix, so $\tilde{Z}(t)$ is invertible over $\kappa[[t]]$. Hence there exists a unique vector $\lambda(t) = (\lambda_1(t), \dots, \lambda_{d'}(t))^\mathsf{T} \in \kappa[[t]]^{d'}$ with $\tilde{Z}(t)\lambda = (q'_{i_1}(t), \dots, q'_{i_{d'}}(t))^\mathsf{T}$. We define $w(t) = q'(t) - Z(t)\lambda(t)$, so that $w_{i_l}(t) = 0$ for $l = 1, \dots, d'$. Its 0th entry is $w_0(t) = q'_0(t) - Z_{0\bullet}(t)\lambda(t)$. Here $q'_0(t) = at^{a-1}u_0(t) + t^a u_0(t)'$ has order exactly $a-1$ (since the coefficient $au_0(0)$ of t^{a-1} is nonzero in characteristic zero), while $Z_{0\bullet}(t)\lambda(t)$ has order at least a, because every entry of row 0 of $Z(t)$ has order at least a (since $q_0(t)$ has order a). Hence $\operatorname{ord} w_0(t) = a-1$. Let c be the minimal order among the entries of $w(t)$; then $c \le a-1$, and $\tau(t) := t^{-c}w(t)$ is a vector over $\kappa[[t]]$ with $\tau(0) \neq 0$ and $\tau_{i_l}(t) = 0$. In row 1, both $q'_1(t)$ and $Z_{1\bullet}(t)\lambda(t)$ have order at least $b-1$, so $\operatorname{ord} w_1(t) \ge b-1$, so $\operatorname{ord} \tau_1(t) \ge b-1-c \ge b-a > 0$. Finally $q'_j(t) = 0$ and $Z_{j\bullet}(t) = 0$ (as $q_j(t) = 1$), so $\tau_j(t) = 0$.

We now show that $q(t), Z_{\bullet 1}(t), \dots, Z_{\bullet d'}(t), \tau(t)$ span all of $\mathcal{C}$: Let $M(t) = (\, q(t), Z_{\bullet 1}(t), \dots, Z_{\bullet d'}(t), \tau(t)\,)$, an $(n+1) \times (d'+2)$ matrix over $\kappa[[t]]$. Its reduction $M(0)$ has full column rank: in row j only $q(0)$ is nonzero ($q_j(0) = 1$); in rows $i_1, \dots, i_{d'}$ the columns $Z_{\bullet k}(0)$ form an identity block while $\tau(0)$ vanishes there; and $\tau(0) \neq 0$, i.e. we have $d'+2$ independent reduced column vectors. This also translates to the column vectors of $M(t)$ themselves: Since some $(d'+2) \times (d'+2)$ minor of $M(0)$ is nonzero, the corresponding minor of $M(t)$ is a unit of $\kappa[[t]]$; hence the columns of $M(t)$ are also linearly independent over $\kappa((t))$, so they span a $(d'+2)$-dimensional subspace there. Furthermore, every column of $M(0)$ has a vanishing entry for the x_1-coordinate: $q_1(0) = 0$ (as $b > 0$), the entries $Z_{1k}(t)$ have order at least $b > 0$, hence $Z_{1k}(0) = 0$, and $\tau_1(0) = 0$. Thus the span of $q(0), Z_{\bullet 1}(0), \dots, Z_{\bullet d'}(0), \tau(0)$ is contained in $\{x_1 = 0\}$ and also has dimension $d'+2$.
To show that the columns of $M(t)$ really span all of $\mathcal{C}$, we must still show that a) the dimensions match and b) the columns actually lie in $\mathcal{C}$. The following Lemma will help us with that.

Lemma 5.10. *Let $m = \operatorname{trdeg}_K F$ and Θ_y, ρ and $\mathcal{C}$ be defined as above.*

(a) $\dim_{\kappa((t))} \Theta_y = m \le d'+1$; hence $\dim_{\kappa((t))} \mathcal{C} = m+1 \le d'+2$.
(b) For every K-derivation δ of $\kappa((t))$ the column $\delta q(t) = (\delta q_i(t))_i$ lies in $\mathcal{C}$; moreover $q(t) \in \mathcal{C}$.

Proof. (a) For a finitely generated field extension F/K in characteristic 0, $\dim_F \Omega_{F/K} = \operatorname{trdeg}_K F = m$ [17, Thm. II.8.6A]. The second fundamental exact sequence [22, 17][...] for $K[x] \to R = K[x]/\mathfrak{p}$ is

$$\mathfrak{p}/\mathfrak{p}^2 \to \Omega_{K[x]/K} \otimes_{K[x]} R \to \Omega_{R/K} \to 0,$$

where $\Omega_{K[x]/K}$ is free on $dx_1, \dots, dx_n$, so the middle term is R^n and the first map sends f to $df(\bar{x})$. Hence $\Omega_{R/K} = R^n / \operatorname{span}_R\{df(\bar{x}) : f \in \mathfrak{p}\}$. For a

multiplicative set $S \subseteq R$, $\Omega_{S^{-1}R/K} = S^{-1}\Omega_{R/K}$. In particular, this means for $F = S^{-1}R$ with $S = R \setminus \{0\}$ $\Omega_{F/K} = F \otimes_R \Omega_{R/K}$ and since localization is exact, applying $F \otimes_R -$ to the description of $\Omega_{R/K}$ above gives

$$\Omega_{F/K} = F^n / \operatorname{span}_F\{df(\bar{x}) : f \in \mathfrak{p}\}.$$

Analogously, applying this for $S = V \setminus \{0\}$ and $F = S^{-1}V$ gives $F \otimes_V \Omega_{V/K} = \Omega_{F/K}$. $\operatorname{span}_F\{df(\bar{x}) : f \in \mathfrak{p}\}$ has therefore dimension $n-m$, i.e. the Jacobian $(\partial_i f(\bar{x}))_{f\in\mathfrak{p}}$ has rank $n-m$ over F. Rank does not change under the field extension $F \subseteq \kappa((t))$, so $\dim \operatorname{span}_{\kappa((t))}\{df_y : f \in \mathfrak{p}\} = n-m$, and therefore $\dim_{\kappa((t))} \Theta_y = \dim_{\kappa((t))} \bigcap_f \ker df_y = n - (n-m) = m$.
The second fundamental exact sequence for $V \to \kappa$ is

$$\mathfrak{m}_V/\mathfrak{m}_V^2 \to \kappa \otimes_V \Omega_{V/K} \to \Omega_{\kappa/K} \to 0.$$

Since $\dim_\kappa \mathfrak{m}_V/\mathfrak{m}_V^2 = 1$ and $\dim_\kappa \Omega_{\kappa/K} = d'$, we get $\dim_\kappa(\kappa \otimes_V \Omega_{V/K}) \le d'+1$. As $\Omega_{V/K}$ is a finitely generated V-module (Lemma 5.8(c)), Nakayama's lemma shows that it is generated by at most $d'+1$ elements. Since $F \otimes_V \Omega_{V/K} = \Omega_{F/K}$, we obtain $m \le d'+1$, hence $\dim \Theta_y \le d'+1$. Finally, ρ maps $\mathcal{C}$ onto Θ_y with one-dimensional kernel, so $\dim_{\kappa((t))} \mathcal{C} = \dim_{\kappa((t))} \Theta_y + 1 \le d'+2$.
(b) Let $f \in \mathfrak{p}$ and $f = \sum_\alpha c_\alpha y^\alpha$. By (5.1), $f(y) = 0$, and δ vanishes on K, so

$$\delta\big(f(y)\big) = \sum_\alpha \Big(\underbrace{y^\alpha\, \delta(c_\alpha)}_{=0} + c_\alpha\, \delta(y^\alpha) \Big) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(y)\, \delta y_i = df_y \cdot \delta y,$$

hence $\delta y \in \Theta_y$. Since $y_i = q_i/q_0$, the quotient rule gives $\rho(\delta q)_i = (\delta q_i\, q_0 - q_i\, \delta q_0)/q_0^2 = \delta(q_i/q_0) = \delta y_i$. Thus $\rho(\delta q) = \delta y \in \Theta_y$, i.e. $\delta q \in \mathcal{C}$. Moreover, $\rho(q)_i = (q_i q_0 - q_i q_0)/q_0^2 = 0$ for all i, so $\rho(q) = 0 \in \Theta_y$ and therefore $q \in \mathcal{C}$. □

The column vectors $q(t)$, $Z_{\bullet k}(t)$ and $q'(t)$ are of the form $\delta q(t)$, where δ is a K-derivation of $\kappa((t))$ (the D_k extended coefficientwise, and d/dt). By Lemma 5.10(b) they lie in $\mathcal{C}$. Since $\tau(t) = t^{-c}(q'(t) - Z(t)\lambda(t))$ and $t^{-c} \in \kappa((t))$, $\tau(t)$ is a $\kappa((t))$-linear combination, hence also lies in $\mathcal{C}$. By Lemma 5.10(a), $\dim \mathcal{C} \le d'+2$, so the $d'+2$ independent columns of $M(t)$ must span $\mathcal{C}$.

Lemma 5.11. *If a row vector $\ell \in \kappa((t))^{n+1}$ annihilates $\mathcal{C}$, then $(\ell_1, \dots, \ell_n) \in \operatorname{span}\{df_y : f \in \mathfrak{p}\}$.*

Proof. Let ℓ annihilate $\mathcal{C}$. Since $q \in \mathcal{C}$, $\ell(q) = 0$, i.e. $\sum_{i\ge1} \ell_i q_i = -\ell_0 q_0$. For $v \in \mathcal{C}$ we have $\ell(v) = 0$, i.e. $\sum_{i\ge1} \ell_i v_i = -\ell_0 v_0$, so

$$\sum_{i\ge1} \ell_i \rho(v)_i = \frac{q_0 \sum_{i\ge1} \ell_i v_i - v_0 \sum_{i\ge1} \ell_i q_i}{q_0^2} = \frac{-q_0 \ell_0 v_0 + v_0 \ell_0 q_0}{q_0^2} = 0.$$

As $\rho(\mathcal{C}) = \Theta_y$, the vector $(\ell_1, \dots, \ell_n)$ annihilates $\Theta_y = \bigcap_f \ker df_y$, hence lies in $\Theta_y^\perp = \operatorname{span}\{df_y : f \in \mathfrak{p}\}$. □

Let $u = (0, 1, 0, \dots, 0)$ be the row vector reading off the x_1-coordinate; since $\operatorname{im} M(0) \subseteq \{x_1 = 0\}$, we know that $uM(0) = 0$. We can lift u to a row vector $\ell(t) \in \kappa[[t]]^{n+1}$ with $\ell(t)M(t) = 0$ and $\ell(0) = u$: Since $M(0)$ has full column rank $d' + 2$, there is a set I of $d' + 2$ row indices for which the square submatrix $M_I(0)$ of $M(0)$ is invertible; then M_I is invertible over $\kappa[[t]]$, since its determinant has a nonzero constant term. Let J be the complementary set of row indices, and define $\ell(t)$ by

$$\ell_J(t) = u_J, \qquad \ell_I(t) = -u_J M_J(t) M_I^{-1}(t),$$

where $M_J(t)$ denotes the submatrix of $M(t)$ with rows in J. Then $\ell(t)M(t) = \ell_I(t)M_I(t) + \ell_J(t)M_J(t) = -u_J M_J(t) + u_J M_J(t) = 0$. Moreover $uM(0) = 0$ gives $u_I = -u_J M_J(0) M_I(0)^{-1}$, hence $\ell(0) = u$.

Since the columns of $M(t)$ span $\mathcal{C}$, $\ell(t)$ annihilates $\mathcal{C}$, so by Lemma 5.10(c) its affine part $\tilde{\ell}(t) = (\ell_1(t), \dots, \ell_n(t))$ lies in $\operatorname{span}\{df_y : f \in \mathfrak{p}\}$, and since $\ell(0) = u$, $\tilde{\ell}(0) = (1, 0, \dots, 0)$.

Finally, let $P \in K[\xi_1, \dots, \xi_n]$ vanish on $N(Y)$. We will show $P(\tilde{\ell}(0)) = 0$: Since $\tilde{\ell}(t) \in \operatorname{span}_{\kappa((t))}\{dg_y : g \in \mathfrak{p}\}$, we can write

$$\tilde{\ell}(t) = \sum_{k=1}^{s} \mu_k(t) \nabla g_k(y), \qquad g_k \in \mathfrak{p},\ \mu_k(t) \in \kappa((t)),$$

With some $s \in \mathbb{N}$. Here $\nabla g(x) = (\partial g/\partial x_1(x), \dots, \partial g/\partial x_n(x))$ denotes the coordinate vector of the covector dg_x. We furthermore define

$$\Phi(x, \mu) = P\Big(\sum_{k=1}^{s} \mu_k \nabla g_k(x)\Big) = \sum_{\alpha} \phi_\alpha(x) \mu^\alpha \in K[x_1, \dots, x_n, \mu_1, \dots, \mu_s],$$

with $\phi_\alpha \in K[x_1, \dots, x_n]$ and $\mu^\alpha = \mu_1^{\alpha_1} \dots \mu_s^{\alpha_s}$ for a multi-index $\alpha \in \mathbb{N}^s$. For a smooth point $y' \in Y_{\mathrm{sm}}$ and $\mu \in K^s$ we have $\sum_k \mu_k dg_k(y') \in N^*_{y'}Y \subseteq N(Y)$, so $\Phi(y', \mu) = 0$. As K is infinite, a polynomial in μ vanishing on all of K^s is zero, so $\phi_\alpha(y') = 0$ for all α. Since Y_{sm} is dense in Y, every ϕ_α vanishes on all of Y, i.e. $\phi_\alpha \in \mathfrak{p} = I(Y)$. By (5.1) (applied to the generic point $y \in \kappa((t))^n$), we therefore have $\phi_\alpha(y) = 0$ for all α. Hence $\Phi(y, \mu(t)) = \sum_\alpha \phi_\alpha(y)\mu^\alpha(t) = \sum_\alpha 0 \cdot \mu^\alpha(t) = 0$ for all $\mu(t) \in \kappa((t))^s$, and in particular

$$P(\tilde{\ell}(t)) = \Phi\big(y, (\mu_1(t), \dots, \mu_s(t))\big) = 0.$$

Now $\tilde{\ell}(t) \in \kappa[[t]]^n$ and P has coefficients in K, so $P(\tilde{\ell}(t)) \in \kappa[[t]]$, and its constant term is $P(\tilde{\ell}(0))$. Since $P(\tilde{\ell}(t)) = 0$, we get $P(\tilde{\ell}(0)) = P(1, 0, \dots, 0) = 0$, which is statement (i).

We have shown that every polynomial P vanishing on $N(Y)$ vanishes at $(1, 0, \dots, 0)$ (statement (i)). By the definition of the Zariski closure, $\operatorname{Dir}(Y)$ is the common zero set of all homogeneous polynomials vanishing on $\{[\xi] : 0 \neq \xi \in N(Y)\}$, equivalently on $N(Y)$. Each such polynomial vanishes at $(1, 0, \dots, 0)$, hence at its class $[dx_1] = [1 : 0 : \dots : 0]$. Therefore $[dx_1] \in \operatorname{Dir}(Y)$. $\square$

Formal comparison: any algebraically closed field of characteristic 0; (i) is the affine-cone form used in Section 5.3

5.3. **Proof of Theorem 5.6.**

Proof. Suppose we have a *nonempty* closed subset $W \subseteq T^*X$ that is fibre-conical and involutive. Suppose further, that $W \cap \pi^{-1}(H) = \varnothing$.
We will give a proof by contradiction, with this rough logic: Suppose $W \neq \varnothing$ and let $Y = \pi(W)$. Since W is by assumption fibre-conical, we start by showing that $Y \times \{0\} = \pi(W) \times \{0\} \subseteq W$ and $Y \cap H = \varnothing$. Using involutivity, we then show that we can translate along W in the direction of the fibres which means that all conormal covectors of every irreducible component Y_0 of Y lie in W. This means P vanishes on all of W, hence on $N(Y_0)$. This gives us the setup of Theorem 5.5, which implies $P(1, 0, \ldots, 0) = 0$, a contradiction to our assumption.

We define the (radical) vanishing ideal $J = I(W) \subseteq K[x, \xi]$ consisting of all polynomials that vanish at every point $(y, \xi) \in W$, and $B = J \cap K[x]$, consisting of the functions $f(x)$ with $f(y) = 0$ for every $(y, \xi) \in W$.
Because W is fibre-conical, the ξ-homogeneous components of any polynomial in J again lie in J: if $g = \sum_d g_d \in J$ and $(y, \xi) \in W$, then $\sum_d a^d g_d(y, \xi) = g(y, a\xi) = 0$ for all $a \in K^\times$, and a polynomial in a with infinitely many roots is zero, so every $g_d(y, \xi) = 0$ and therefore $g_d \in J$. Let $g \in J$ be written in its ξ-homogeneous components form. In particular, $g_0 \in J$. Since g_0 has ξ-degree 0, it is a polynomial in x alone, so $g_0 \in B$. Let $Y = V(B)$. For $y \in Y$ we therefore have $g(y, 0) = g_0(y) = 0$, so every $g \in J$ vanishes at $(y, 0)$, i.e. $(y, 0) \in V(J)$. Since W is closed, $V(J) = V(I(W)) = W$, hence $(y, 0) \in W$, or $Y \times \{0\} \subseteq W$. Furthermore, Y is the projection of W and is therefore per definition of Y closed: $\pi(W) \subseteq Y$ since for $(y', \xi') \in W$ and $f \in B \subseteq J$, we have $f(y') = f(y', \xi') = 0$ so $y' \in V(B) = Y$, and $Y \subseteq \pi(W)$, since if $y \in Y$, then also $(y, 0) \in W$, so $y = \pi(y, 0) \in \pi(W)$.
Lastly, assume there exists a $y \in Y \cap H$, then $(y, 0) \in W$ and $\pi(y, 0) = y \in H$, which would put $(y, 0) \in W \cap \pi^{-1}(H)$, which is wrong by our assumption. Therefore, $Y \cap H = \varnothing$.

Let $f \in B$. Then $f \circ \pi \in J$. The Hamiltonian derivation of f applied to a $a \in J$ is

$$H_f g = \sum_i \frac{\partial f}{\partial x_i} \frac{\partial g}{\partial \xi_i} = \{f, g\}$$

since the second term of the Poisson bracket vanishes because f does not depend on ξ. Since $f, g \in J$ and $\{J, J\} \subseteq J$ by the involutivity of W, it follows that $H_f g \in J$ and by induction also $H_f^m g \in J$. For a fixed $y \in Y$ and $g \in J$, we define the polynomial $\varphi(t) = g(y, t\, df_y) \in K[t]$. Its derivative is $\varphi'(t) = \sum_i \partial_{x_i} f(y) \frac{\partial g}{\partial \xi_i}(y, t df_y) = H_f g(y, t df_y)$, so by iteration, $\varphi^{(m)}(t) = H_f^m g(y, t df_y)$ for $m \geq 0$. Evaluating at $t = 0$, $\varphi^{(m)}(0) = H_f^m g(y, 0)$.

Since we have established that $(y, 0) \in W$ and $H_f^m g \in J$, and all polynomials in J vanish on W, we have $\varphi^{(m)}(0) = 0$ for all $m \geq 0$ – so $\varphi(t)$ is a polynomial it t all of whose derivatives vanish at $t = 0$. If we write $\varphi(t) = \sum_m c_m t^m$, then $\varphi^{(m)}(0) = m! c_m$ and since $m! \neq 0$ because K has characteristic zero, $c_m = 0$ for all m, and therefore, $\varphi(t) = 0$ everywhere. Remembering the definition of $\phi = g(y, t\, df_y) = 0$, so $(y, t\, df_y) \in V(J) = W$. More generally, since any linear combination of elements in B is again in B, we can apply this result to $f = \sum_a c_a f_a \in B$ and evaluating at $t = 1$, it follows that

$$(y, df_y) = (y, d(\sum_a c_a f_a)_y) = (y, \sum_a c_a d(f_a)_y) \in W,$$

so the whole linear span $\{y\} \times \operatorname{span}\{df_y : f \in B\}$ is a subset of W.

Since $V(B) = Y = \pi(W) \neq \varnothing$, $B \neq K[x]$, so B must have at least one minimal prime. The ideal B is radical (by construction since it is a vanishing ideal) in the Noetherian ring $K[x]$, so B has only finitely many minimal primes $\mathfrak{q}_1, \ldots, \mathfrak{q}_r$ and $B = \bigcap_i \mathfrak{q}_i$. We choose $\mathfrak{q} = \mathfrak{q}_1$ and set $Y_0 = V(\mathfrak{q})$. Since $\mathfrak{q}$ is prime, $1 \notin \mathfrak{q}$, so $Y_0 \neq \varnothing$ by the Nullstellensatz (Theorem 2.8); moreover Y_0 is closed and irreducible since $\mathfrak{q}$ is prime [17, Cor. I.1.4], and $Y_0 \subseteq Y$, hence $Y_0 \cap H = \varnothing$, as we have already established $Y \cap H = \varnothing$.
Furthermore, if $r = 1$, we choose $s = 1$, then $sh \in \mathfrak{q} = B$, and if $r \geq 2$, choose for every $i \geq 2$ elemens $s_i \in \mathfrak{q}_i \setminus \mathfrak{q}$. Then $s = s_2 s_3 \ldots s_r$ does not lie in $\mathfrak{q}$ since no factor lies in $\mathfrak{q}$, but does lie in every other $\mathfrak{q}_i$ since they are ideals and one of the factors lies in them. On the other hand, if $h \in \mathfrak{q}$, then $sh \in \mathfrak{q}$ since $h \in \mathfrak{q}$, and $sh \in \mathfrak{q}_i$ since $s \in \mathfrak{q}_i$ for all $i \geq 2$. Therefore, $sh \in \bigcap_i \mathfrak{q}_i = B$, i.e. $s\mathfrak{q} \subseteq B$. We now fix such a $h \in \mathfrak{q}$, and define $Y_0 = V(\mathfrak{q})$. For a $y \in Y_0 \cap \{s \neq 0\}$, $(y, dh_y) \in W$: Since $sh \in B$, and $d(sh)_y = s(y)dh_y + h(y)ds_y = s(y)dh_y$ since $h(y) = 0$, so $(y, s(y)dh_y) \in W$ and since $s(y) \neq 0$, also $(y, dh_y) \in W$. For $g \in J$, the polynomial $G(x) = g(x, dh_x)$ therefore vanishes on $Y_0 \cap \{s \neq 0\}$, and sG vanishes on all of Y_0, so $sG \in \mathfrak{q}$ and therefore $G \in \mathfrak{q}$, which means that G vanishes on all of Y_0. Since this is true for every $g \in J$, we have $g(y, dh_y = G(y) = 0$ for all $g \in J$, so $(y, dh_y) \in V(J) = W$. As before, we can argue that since $\mathfrak{q}$ is an ideal, it is closed under K-linear combinations, so for $h_1, \ldots, h_k \in \mathfrak{q}$ and $c_a \in K$, also $(y, \sum_a c_a (dh_a)_y) \in W$, or put differently, $\{y\} \times \operatorname{span}\{dh_y : h \in \mathfrak{q}\} \subseteq W$ for all $y \in Y_0$. At a smooth point y of Y_0, by Definition **??**, the conormal space is

$$N_y^* Y_0 = \operatorname{span}\{df_y : f \in \mathfrak{q}\}$$

since $\mathfrak{q}$ is prime and hence radical, and so $\mathfrak{q} = I(Y_0)$. We therefore have that $N_y^* Y_0 \subseteq \{y\} \times W$ for every smooth $y \in Y_0$, and hence $N(Y_0) \subseteq W$.
By assumption, P vanishes on W and hence on $N(Y_0)$. Because we have established, that Y_0 is closed, irreducible, and $Y_0 \cap H = \varnothing$, Y_0 therefore fulfills all requirements of Theorem 5.5, so $P(1, 0, \ldots, 0) = 0$, contradicting our initial assumption. Therefore, W must be empty. □

6. Proof of Theorem 1.1

Proof. $Q_\bullet$ is a good filtration because Q is cyclic. Let $\bar{q}$ be the class of q in $\operatorname{gr} Q$. We know that $q\,\tilde{d} = [\tilde{d}] = 0$ in Q (because $\tilde{d} \in I$) and $\tilde{d}$ has order N (Lemma 3.5), so $q\tilde{d} \in Q_N$. Therefore $\bar{q} \cdot P$, its image in Q_N/Q_{N-1}, must also be zero, and hence P annihilates the class $\bar{q}$ of q in $\operatorname{gr} Q$. Since $\operatorname{gr} Q$ is generated over $k[x,\xi]$ by $\bar{q}$, we get $P \in \operatorname{Ann}_{k[x,\xi]} \operatorname{gr} Q$. □

Proposition 6.1. $\operatorname{Char}(Q) \cap \pi^{-1}(H) = \varnothing$.

Proof. Write $M = \operatorname{gr} Q$. Since M is finite over $k[x,\xi]$, $\operatorname{Supp}_{k[x,\xi]}(M/x_1M) = \operatorname{Supp}_{k[x,\xi]}(M) \cap V(x_1)$ [37, Tag 00L3], where $V(x_1)$ is the set of primes of $k[x,\xi]$ containing x_1. By Corollary 4.11, $M/x_1M = 0$, so its support is empty, hence $\operatorname{Supp}_{k[x,\xi]}(M) \cap V(x_1) = \varnothing$. By definition $\operatorname{Supp}_{k[x,\xi]}(M) = \operatorname{Char}(Q)$, and $V(x_1) = \pi^{-1}(H)$ is the preimage of $H = \{x_1 = 0\}$ under the projection π. Therefore $\operatorname{Char}(Q) \cap \pi^{-1}(H) = \varnothing$. □

Formal comparison: stated for the support transported along $\xi \mapsto -\xi$, which fixes x_1; transported back verbatim

Lemma 6.2. *Let K/k be a field extension and put $\mathcal{I}_K = \tilde{d}A_n(K) + x_1^N\tilde{d}A_n(K)$, $Q_K = Q \otimes_k K = A_n(K)/\mathcal{I}_K$, with the order filtration $Q_{K,i} = q\,F_iA_n(K)$. Then*

$$\operatorname{gr} Q_K \cong \operatorname{gr} Q \otimes_k K, \qquad \operatorname{Ann}_{K[x,\xi]} \operatorname{gr} Q_K = \big(\operatorname{Ann}_{k[x,\xi]} \operatorname{gr} Q\big)\,K[x,\xi],$$

and $Q = 0 \iff Q_K = 0$. In particular, $\operatorname{Char}(Q_K)$ is the preimage of $\operatorname{Char}(Q)$ under $\operatorname{Spec} K[x,\xi] \to \operatorname{Spec} k[x,\xi]$.

Proof. By the PBW basis, $F_iA_n(K) = F_iA_n(k) \otimes_k K$ and $\mathcal{I}_K = \mathcal{I} \otimes_k K$, so $Q_{K,i} = Q_i \otimes_k K$. As K is flat over k, $Q_{K,i}/Q_{K,i-1} = (Q_i/Q_{i-1}) \otimes_k K$, and summing over i gives $\operatorname{gr} Q_K = \operatorname{gr} Q \otimes_k K$. The module $\operatorname{gr} Q$ is finitely generated (cyclic, generated by the class of q) over the Noetherian ring $k[x,\xi]$, and $k[x,\xi] \to K[x,\xi]$ is flat, so the annihilator of the finite module commutes with this base change [37, Tag 05DT]. Finally, $Q \otimes_k K = 0$ implies $Q = 0$, because K is a nonzero free k-module. □

Let $d \in A_n(k)$ be nonzero of Bernstein degree N and let $\tilde{d} = c^{-1}\Phi_g(d)$ be its monic normal form (Remark 3.3). If $n = 0$, then $A_0(k) = k$ and every nonzero d is a unit anyways, so let $n \geq 1$. As in Section 4 we define

$$\mathcal{I} = \tilde{d}A_n(k) + x_1^N\tilde{d}A_n(k), \qquad Q = A_n(k)/\mathcal{I}.$$

It suffices to show $Q = 0$: then $1 \in \mathcal{I}$, i.e. there are $R, S \in A_n(k)$ with $1 = \tilde{d}R + x_1^N\tilde{d}S$.
Let $K = \overline{k}$, and let

$$W = \operatorname{Char}(Q_K) = V\big(\operatorname{Ann}_{K[x,\xi]} \operatorname{gr} Q_K\big) \subseteq K^n \times K^n,$$

with vanishing ideal $I(W) = \sqrt{\operatorname{Ann} \operatorname{gr} Q_K}$ by the Nullstellensatz (Theorem 2.8). We check the hypotheses of Theorem 5.6 for W and the polynomial $P = \sigma_N^{\mathrm{ord}}(\tilde{d})$ of Section 3:

- W is Zariski closed, since it is the zero set of an ideal of polynomials.
- The module $\operatorname{gr} Q_K$ is graded by the order filtration, which assigns degree 0 to the x_i and degree 1 to the ξ_i; so $\operatorname{Ann} \operatorname{gr} Q_K$ is homogeneous in ξ. Hence if $(y, \xi) \in W$ and $g \in \operatorname{Ann} \operatorname{gr} Q_K$, then $g(y, t\xi) = \sum_d t^d g_d(y, \xi) = 0$ for every $t \in K^\times$, where g_d denotes the ξ-homogeneous component of degree d of g. So $(y, t\xi) \in W$, i.e. W is fibre-conical.
- By Gabber's theorem (Theorem 2.7) $I(W)$ is involutive.
- By eq. (3.5), $P \in \operatorname{Ann}_{k[x,\xi]} \operatorname{gr} Q$, hence $P \in \operatorname{Ann}_{K[x,\xi]} \operatorname{gr} Q_K$ by Lemma 6.2, so P vanishes on W.
- By eq. (3.4), $P(1, 0, \ldots, 0) = 1 \neq 0$.
- $W = \operatorname{Char}(Q_K)$ is the preimage of $\operatorname{Char}(Q)$ under $\operatorname{Spec} K[x, \xi] \to \operatorname{Spec} k[x, \xi]$ (Lemma 6.2), and $\pi^{-1}(H) = V(x_1)$ over K is the preimage of $\pi^{-1}(H) = V(x_1)$ over k. By Proposition 6.1, $\operatorname{Char}(Q) \cap \pi^{-1}(H) = \varnothing$, and the preimage of an empty set is also empty, therefore $W \cap \pi^{-1}(H) = \varnothing$.

By Theorem 5.6, $W = \varnothing$. As K is algebraically closed, by the Nullstellensatz (Theorem 2.8) $1 \in \operatorname{Ann}_{K[x,\xi]} \operatorname{gr} Q_K$, so $\operatorname{gr} Q_K = 0$. Since the filtration of Q_K is exhaustive with $Q_{K,-1} = 0$, induction on m implies $Q_{K,m} = 0$ for all m, so $Q_K = 0$, and $Q = 0$ by Lemma 6.2. Therefore, $A_n(k) = \tilde{d} A_n(k) + x_1^N \tilde{d} A_n(k)$, or equivalently $1 = \tilde{d} R + x_1^N \tilde{d} S$ with $R, S \in A_n(k)$.
Lastly, we apply the inverse of the automorphism Φ_g of Remark 3.3. Since $\tilde{d} = c^{-1} \Phi_g(d)$ with $c \in k^\times$ central, this gives

$$1 = d\big(c^{-1} \Phi_g^{-1}(R)\big) + F\, d\big(c^{-1} \Phi_g^{-1}(S)\big), \qquad F = \Phi_g^{-1}(x_1)^N,$$

which proves Theorem 1.1. □

Formal comparison: stated for the canonical ideal; the argument uses nothing else

Lastly, as Stafford already showed, from $1 = dR + FdS$ follows cyclicity of the torsion right module:

Corollary 6.3. *Every finitely generated torsion right $A_n(k)$-module is cyclic.*

Proof. Suppose for every $d \neq 0 \in A_n$, there exists $F, R, S \in A_n$, such that $1 = dR + FdS$. Let us denote a torsion module by T. Stafford's theorem (Theorem 3.7 [30, p. 438]) established, that every finitely generated torsion module can be generated by 2 generators, so let us denote the generators of T with t_1 and t_2. We can define a homomorphism

$$\phi : A_n \oplus A_n \to T \quad \phi(r, s) \mapsto t_1 r + t_2 s.$$

This map is surjective, since t_1, t_2 generate all of T. By the first isomorphism theorem (for $\phi : M \to N$, $im(\phi) \cong M/ker(\phi)$), it follows

$$T \cong (A_n \oplus A_n)/M$$

with $M = ker(\phi)$.
T is torsion and therefore $\forall t \in T : \exists m \in A_n \backslash \{0\} : tm = 0$. Specifically, let us denote with d_1 and d_2 the annihilators of t_1 and t_2, i.e.

$$t_1 d_1 = t_2 d_2 = 0.$$

A_n is a right Ore domain, which means it has the property that for any nonzero d_1, d_2 there exists elements $r_1, r_2 \in A_n$ such that $d_1 r_1 = d_2 r_2 \neq 0$. Defining therefore $d := d_1 r_1 = d_2 r_2$ yields us a common annihilator of t_1 and t_2.
$(d, 0)$ and $(0, d)$ are elements in $M = ker(\phi)$, since $\phi(d, 0) = t_1 d = 0$ and $\phi(0, d) = t_2 d = 0$.
Because $(0, d), (d, 0) \in M$, also $(d, 0)S + (0, d)R = (dS, dR) = (dS, 1 - FdS) =: (a, 1 - Fa) \in M$, i.e. $(a, 1 - Fa)$ is an element of M.
On the other hand, $A_n \oplus A_n$ can be generated by $(a, 1 - Fa)$ and $(1, -F)$. We can check this by finding u, v such that for all $(x, y) \in A_n \oplus A_n$ we can write

$$(x, y) = (a, 1 - Fa)u + (1, -F)v.$$

Suitable choices are $u = y + Fx$ and $v = x - a(y + Fx)$.
Lastly, since $A_n \oplus A_n$ can be generated by $(a, 1 - Fa)$ and $(1, -F)$ and $(a, 1 - Fa)$ is an element of M, it follows that $T \cong (A_n \oplus A_n)/M$ can be generated by $\phi(1, -F)$, i.e. T is cyclic.
Moreover, Stafford defined a module T to be stably cyclic, if, given $r \geq 1$ and $a_1, ...a_{r+1} \in T$ that generate T, there exist $f_i \in A_n$ such that $(a_i + a_{r+1} f_i)$ $i = 1, ...r$ also generate T [30, p. 435]. Since t_1, t_2 were arbitrary generators, for $r = 1$ we can set $a_1 = t_1, a_2 = t_2$ and choose $f_1 = -F$, then $\phi(1, -F) = t_1 - t_2 F$ generates T. For $r \geq 2$ we simply note that by [30, Theorem 3.7], T is already proven to be stably two-generated, i.e. the case reduces again to $r = 1$. By choosing $a_1 = (1, 0), a_2 = (0, -1), f_0 = F$, we therefore see that our T is stably cyclic. Since t_1, t_2 above is an arbitrary generating pair of T, this settles the case $r = 1$ for every pair; for $r \geq 2$, T is stably two-generated by [30, Theorem 3.7]. □

AI comment CYC-01. Proposed completion — author acceptance pending. The conclusion is correct. Stable cyclicity is required for every generating tuple $a_1, \ldots, a_{r+1}$, $r \geq 1$; the added sentence covers all pairs and $r \geq 2$, following Stafford's own argument on p. 438.

7. Generalizations and open questions

The formalization of Section 1.2 also proves the following consequences, which are not used here and have no printed proof: the identity in every right Ore localization of $A_n(k)$; the left-handed identity $1 = Rd + SdF$, via the formal adjoint; the identity for finite-order differential operators on localizations of $k[x_1, \ldots, x_n]$, including principal opens, Laurent-polynomial coefficients and the fraction field; and, independently of the geometric argument, the identity

with $F = x^r$ for $d = p^r - \sum_j a_j x^{m_j}$, $r \geq 1$, in any k-algebra containing elements x, p with $px = xp + 1$, provided each a_j commutes with x and p. Two further results, also proved in Lean, give the identity $1 = dR + FdS$ in larger classes of rings, without the choice of F in Theorem 1.1. Let k be a field of characteristic zero.

(a) Let A be a smooth finitely generated k-algebra that is a domain, and let $D_k(A)$ be the ring of k-linear differential operators of finite order on A. For every nonzero $d \in D_k(A)$ there are $F, R, S \in D_k(A)$ with $1 = dR + FdS$ [3, 4]. For $A = k[x_1, \ldots, x_n]$ this is the identity in $A_n(k)$.

(b) Let U be an infinite-dimensional simple k-algebra with an exhaustive filtration $k = F_0U \subseteq F_1U \subseteq \cdots$ by finite-dimensional subspaces, $F_iU \cdot F_jU \subseteq F_{i+j}U$, whose associated graded ring is a commutative domain finitely generated over k. For all nonzero $d, a \in U$ and every $B \in U$ there are $H, R, S \in U$ with $1 = dR + (B + Ha)dS$ [5]. The Bernstein filtration makes $A_n(k)$ such an algebra, and $B = 0$, $a = 1$ gives $1 = dR + HdS$.

The proof of Theorem 1.1 shows $Q = 0$ without producing R and S. ~~Effective bounds for $\deg_B R$ and $\deg_B S$ in terms of $\deg_B d$, and an algorithm in the spirit of the constructive two-generator methods, remain open, as do the corresponding questions for (a) and (b).~~ Over a computable field, the fixed choice of F nevertheless gives a terminating algorithm: enumerate pairs (g, D) of linear symplectic changes of generators and degree bounds, and solve the linear system $1 = dR + FdS$ with $F = \Phi_g^{-1}(x_1)^{\deg_B d}$ for R, S of Bernstein degree at most D. Theorem 1.1 guarantees that some pair succeeds. Bounds for $\deg_B R$ and $\deg_B S$ in terms of $\deg_B d$, and with them an efficient algorithm, remain open. In (a) and (b), F is not fixed, so this search does not apply directly.

AI comment ALG-05. Proposed correction — author acceptance pending. The earlier text called any algorithm open. The fixed-source form makes the search above terminate; what remains open is efficiency and degree bounds.

AI comment FV-03. Placeholder. Statement (b) is formalized without `sorry` and uses only the three standard axioms; its citation is a placeholder until the manuscript and archive exist. This section replaces the former Outlook and collects the extensions previously in the introduction.

References

1. Christopher Albert, *Stafford's Conjecture 3.8 for Weyl algebras: Lean 4 formalization v1.3.1*, Zenodo software release v1.3.1, DOI: https://doi.org/10.5281/zenodo.23127367.
2. ———, *AlgebraicAnalysis v0.3.3*, Zenodo software release v0.3.3, 2026, Source revision bbbbf3fc358ca8100b158cec4cf47f336ab70163; DOI: https://doi.org/10.5281/zenodo.23104842.
3. ———, *Global Stafford*, Palomar Registry, https://palomar-registry.org/entry?id=PALOMAR-2026-09-09-000001&version=2, 2026, PALOMAR-2026-09-09-000001, version 2.
4. ———, *Global Stafford: Lean 4 formalization v1.0.6*, Zenodo software release v1.0.6, 2026, DOI: https://doi.org/10.5281/zenodo.22669444.
5. ———, *Same-divisor identities for simple filtered algebras*, 2026, In preparation.
6. ———, *Stafford 3.8*, Palomar, PALOMAR-2026-09-05-000007 v3, https://palomar-registry.org/entry?id=PALOMAR-2026-09-05-000007&version=3, 2026.
7. ———, *Stafford's conjecture 3.8: paper and Lean proof comparison v0.2.2*, Zenodo supplementary release v0.2.2, 2026, DOI: https://doi.org/10.5281/zenodo.23132562.
8. Gwyn Bellamy, *Module structure of Weyl algebras*, Journal of the London Mathematical Society **113** (2026), no. 1, e70373.
9. Jan-Erik Björk, *Rings of differential operators*, North-Holland Mathematical Library, vol. 21, North-Holland, Amsterdam, 1979.
10. Napoleon Caro and Daniel Levcovitz, *On a theorem of Stafford*, Cadernos de Matemática **11** (2010), 63–70, Preprint at https://arxiv.org/abs/1005.4427; journal PDF at the authors' university repository.
11. Napoleon Caro-Tuesta and Daniel Levcovitz, *Module structure of certain rings of differential operators*, Algebras and Representation Theory **23** (2020), 1637–1657.
12. Frédéric Chyzak, Alban Quadrat, and Daniel Robertz, *OreModules: a symbolic package for the study of multidimensional linear systems*, Applications of Time Delay Systems, Lecture Notes in Control and Information Sciences, vol. 352, Springer, 2007, pp. 233–264.
13. P. M. Cohn, *Algebra*, 2nd ed., vol. 1, John Wiley & Sons, Chichester, 1982.
14. Severino C Coutinho, *A primer of algebraic d-modules*, no. 33, Cambridge University Press, 1995.
15. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28 (Cham), Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
16. Ofer Gabber, *The integrability of the characteristic variety*, American Journal of Mathematics **103** (1981), no. 3, 445–468.
17. Robin Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. 52, Springer, New York, 1977.
18. Andre Hillebrand and Wiland Schmale, *Towards an effective version of a theorem of Stafford*, Journal of Symbolic Computation **32** (2001), no. 6, 699–716.
19. Ryoshi Hotta, Kiyoshi Takeuchi, and Toshiyuki Tanisaki, *D-modules, perverse sheaves, and representation theory*, Progress in Mathematics, vol. 236, Birkhäuser, Boston, 2008.
20. LeanProver contributors, *Comparator: statement and proof comparison for Lean*, Project-pinned commit 575674928e239f5bc452aab72d1dd7b0f1326494. Source: https://github.com/leanprover/comparator/tree/575674928e239f5bc452aab72d1dd7b0f1326494.
21. Anton Leykin, *Algorithmic proofs of two theorems of Stafford*, Journal of Symbolic Computation **38** (2004), no. 6, 1535–1550.

22. Hideyuki Matsumura, *Commutative ring theory*, Cambridge Studies in Advanced Mathematics, vol. 8, Cambridge University Press, Cambridge, 1986, Translated from the Japanese by M. Reid.
23. Tadasi Nakayama, *A remark on finitely generated modules*, Nagoya Mathematical Journal **3** (1951), 139–140.
24. *NanoDa: an independent kernel checker for Lean*, Project-pinned commit 68d5ca9db226849b41a6fff59d796ff19d0a8840. Source: https://github.com/ammkrn/nanoda_lib/tree/68d5ca9db226849b41a6fff59d796ff19d0a8840.
25. Palomar Registry, *About the Palomar registry*, Official documentation, accessed 2026-09-30: https://palomar-registry.org/about.html.
26. Alban Quadrat and Daniel Robertz, *Computation of bases of free modules over the Weyl algebras*, Journal of Symbolic Computation **42** (2007), no. 11–12, 1113–1141.
27. ———, *A constructive study of the module structure of rings of partial differential operators*, Acta Applicandae Mathematicae **133** (2014), no. 1, 187–234.
28. Christian Schnell, *D-modules*, 2019, Lecture notes.
29. Jyoti Singh and Shiv Datt Kumar, *On the involutivity of the characteristic variety*, Communications in Algebra **42** (2014), no. 8, 3607–3618.
30. J. T. Stafford, *Module structure of Weyl algebras*, Journal of the London Mathematical Society. Second Series **18** (1978), no. 3, 429–442.
31. ———, *Generating modules efficiently: algebraic K-theory for noncommutative noetherian rings*, Journal of Algebra **69** (1981), no. 2, 312–346.
32. ———, *On the stable range of right noetherian rings*, Bulletin of the London Mathematical Society **13** (1981), no. 1, 39–41.
33. ———, *Generating modules efficiently over noncommutative rings*, Séminaire d'Algèbre Paul Dubreil et Marie-Paule Malliavin, Proceedings. Paris 1981 (34ème Année), Lecture Notes in Mathematics, vol. 924, Springer, Berlin, 1982, pp. 72–88.
34. ———, *Corrigendum*, Journal of Algebra **82** (1983), no. 1, 294–296, Corrigendum to Stafford's 1981 article on generating modules efficiently.
35. ———, *Stably free, projective right ideals*, Compositio Mathematica **54** (1985), no. 1, 63–78, Available at https://www.numdam.org/item/CM_1985__54_1_63_0/.
36. The mathlib Community, *The Lean mathematical library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (New York), ACM, 2020, pp. 367–381.
37. The Stacks Project Authors, *Stacks project*, 2026.

Appendix A. AI-assisted research and formalization

Proof discovery, counterexample searches, review, and formalization were assisted by agentic AI tools. The work began with cyclic-extension computations in Macaulay2 and Maple. Attempts to construct universal cofactors directly produced special cases and reductions but eventually stalled. The decisive change was to study the canonical quotient $Q = A_n/(dA_n + x^N dA_n)$: Euler surjectivity and monicity constrained its characteristic support, while Gabber's involutivity and asymptotic conormal geometry forced that support to be empty. This turned cofactor selection into a quotient-vanishing problem. Full Lean formalization was deferred until the paper argument was complete, although small formal checks continued throughout. In the final formal development, support avoidance over the coordinate hyperplane was proved by a localized filtered Koszul length contradiction.

The workflow separated proposed ideas from verified claims. One controller maintained the authoritative proof state, while parallel workers explored distinct mechanisms and returned evidence with an explicit scope. Independent reviewers checked fixed revisions for quantifiers, sidedness, circularity, and missing hypotheses. The project protocols were based on ~~a `math-frontier` agent skill~~ agent instructions developed for this purpose: freeze the claim, identify its first unresolved dependency, and specify how a route could fail. Conceptual prompts requested a small number of distinct mechanisms; paper-only tasks sought complete arguments before implementation. A condensed representative prompt, adapted from the archived protocols, is:

> Work on one precisely stated unresolved claim. Propose at most three distinct mechanisms, each with an intermediate lemma, a noncircular justification, and an adversarial test. Preserve the quantifiers and right-sided cofactors. For a paper task, supply a complete argument; for formalization, prove the required hypotheses without adding axioms or using `sorry`. Return evidence and the first unsupported bridge; do not promote a conditional result to a theorem.

Research and review sessions used GPT-5.5, GPT-5.6 Luna and Sol, GPT-6 Astra, and Claude Fable 5, Haiku 4.5 and Opus 5. The project protocols assigned routine implementation and replay to Luna and conceptual consultation to Sol; independent reports also record Opus reviews. Subsequent manuscript, proof-map, build and release work used Fable 5.1, Opus 5, Sonnet 5 and GPT-6 Luna. These records establish participation and task roles, without assigning a particular mathematical idea to one model. The final formal campaign, which connected the printed asymptotic conormal argument to the terminal theorem and prepared the releases, used GPT-6 Luna workers and escalated failed bounded tasks to GPT-6.1 Sol.

AI comment AI-01. Chris — please confirm. The campaign ledger records an earlier serial Pi/Qwen run that ported three archived helper lemmas before the final campaign, but no exact model identifier. Name the Qwen model here if it should be disclosed.

The completed proof uses Lean 4 and Mathlib and pins AlgebraicAnalysis ~~v0.3.0~~ v0.3.3 [15, 36, 2]~~; the later library archive is v0.3.3~~. Formalization distinguished theorems proving required inputs from conditional theorems merely consuming them. Statement comparison and independent NanoDa and Lean kernel checks verified the submitted declarations [20, 24, 25]; their proofs use only `propext`, `Classical.choice`, and `Quot.sound`. ~~The Stafford38 and Global Stafford headline proofs~~ Theorem 1.1 and generalization (a) of Section 7 are registered separately in Palomar [6, 3]; their Lean developments are archived [1, 4].

Graz University of Technology